

Statistics for Data Science 1

Course Revision Textbook & NPTEL Lecture Companion

IIT Madras BS in Data Science and Applications

Prepared for Comprehensive Revision & Concept Mastery

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Introduction

Welcome to the **Statistics for Data Science 1 (SDS1)** Revision Textbook. This guide covers every topic from the 12-week IIT Madras foundational course, building from descriptive summaries of raw data all the way to classical probability distributions used in machine learning models.

Structure of this Guide

- **Chapters 1–4 (Quiz 1 Content):** Data types, scales, descriptive statistics for categorical and numerical data, measures of central tendency, dispersion, and association between variables.
- **Chapters 5–8 (Quiz 2 Content):** Counting principles, permutations, combinations, sample spaces, probability axioms, conditional probability, independence, and Bayes' Theorem.
- **Chapters 9–12 (End-Term Content):** Random variables, PMF/CDF, expectation, variance, Binomial, Poisson, and continuous distributions (Uniform, Exponential).

Box Colour Legend

- **Teal** — Definitions and core terminology.
- **Indigo** — Theorems, rules, and important properties.
- **Violet** — Worked examples with step-by-step solutions.
- **Amber** — Key formulas to memorise.
- **Red card** — Video resources (IITM + supplementary).

Chapter 1

Introduction and Types of Data

Statistics is the science of collecting, organising, analysing, and interpreting data to make decisions. In the context of the IITM BS Degree, statistics forms the foundational grammar of Data Science and Machine Learning. Before algorithms can learn from data, we must understand how to classify, measure, and describe that data.

1.1 What is Statistics?

At its core, statistics helps us understand the world by managing uncertainty and variation in data. We divide the field into two major branches.

Definition: Descriptive vs. Inferential Statistics

- **Descriptive Statistics:** Methods for *summarising and presenting* data from a dataset you already have. This involves organizing data into tables, graphs (like histograms or scatter plots), and calculating summary numbers (like the average or variance).
- **Inferential Statistics:** Methods for drawing *conclusions about an entire population* based on a smaller sample drawn from it. This involves probability, hypothesis testing, and building predictive models.

Definition: Population and Sample

- **Population:** The complete, exhaustive set of all individuals or items of interest. Its size is typically denoted by N .
- **Sample:** A subset of the population that is actually observed or measured. Its size is denoted by n . A **representative sample** accurately reflects the characteristics of the population.
- **Parameter:** A numerical summary of a *population* (e.g., population mean μ , which is usually unknown).
- **Statistic:** A numerical summary of a *sample* (e.g., sample mean $\bar{x}$, which is calculated from the data).

Data Science Relevance: In machine learning, your training dataset is just a *sample*. Your goal is to train a model that generalizes to the *population* of all future unseen data. If your sample is biased, your model's inference will be completely flawed (a phenomenon known as "garbage in,

garbage out").

Video Resource: IITM BS Lecture: W0 Introduction to Statistics for Data Science 

Course introduction laying out how statistical thinking ties directly into data science applications.

Video Resource: IITM BS Lecture: W1_L1 Introduction & Basic Definitions 

Official lecture defining Descriptive vs. Inferential statistics, populations, and samples.

1.2 Types of Data

Before analyzing data, we must classify what type of data we are looking at. A **variable** is any characteristic that can differ from individual to individual.

Definition: Variable Classification

- **Categorical (Qualitative):** Values are labels or categories describing attributes.
 - *Nominal* No natural or logical ordering. (Example: Blood type (A, B, AB, O), City name, Gender).
 - *Ordinal* A natural ordering exists, but the physical differences between values are not necessarily equal or meaningful. (Example: Customer satisfaction rating (Poor, Fair, Good, Excellent), T-shirt sizes (S, M, L)).
- **Numerical (Quantitative):** Values are actual numbers representing quantities on which arithmetic is meaningful.
 - *Discrete* Countable values with distinct gaps between them. Usually results from counting. (Example: Number of children in a family (0, 1, 2...), number of cars passing a toll).
 - *Continuous* Can take any value within a given interval. Usually results from measuring. (Example: Height, weight, temperature, exact time).

Video Resource: IITM BS Lecture: W1_L3 Classification of Data 

Official lecture breaking down Nominal, Ordinal, Discrete, and Continuous data with extensive examples.

1.3 Scales of Measurement

The **scale of measurement** is a critical concept that strictly determines which statistical operations (mean, median, mode) and which ML algorithms are mathematically valid to use.

Theorem/Rule: Four Scales of Measurement

Scale	Order?	Equal gaps?	True zero?	Example
Nominal	No	No	No	Pincodes, Colors
Ordinal	Yes	No	No	Movie Ratings (1–5 stars)
Interval	Yes	Yes	No	Temperature (°C), Year (e.g. 2024)
Ratio	Yes	Yes	Yes	Height, Weight, Salary, Age

Note: For Interval scales, a zero is arbitrary (0°C does not mean "no temperature"). For Ratio scales, a zero represents the total absence of the quantity, meaning statements like "twice as heavy" or "half as expensive" are mathematically valid.

Video Resource: IITM BS Lecture: W1_L4 Scales of Measurement



Official lecture detailing the operations allowed on Nominal, Ordinal, Interval, and Ratio scales.

1.4 Cross-Sectional vs. Time Series Data

Data is also classified by how it is collected with respect to time.

Definition: Data Structures by Time

- **Cross-Sectional Data:** Observations taken on multiple subjects at a *single point in time*. (Example: A survey of 500 students' heights taken today, or current stock prices of 50 companies).
- **Time Series Data:** Observations taken on a *single subject over multiple time periods*. (Example: The monthly GDP of India from 2010 to 2025, or a single patient's daily blood pressure).
- **Panel (Longitudinal) Data:** A combination of both; observing multiple subjects over multiple time periods.

Supplementary Lecture Videos

- [W1_L2: Introduction & types of data - understanding data](#)

1.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Concepts:** A medical researcher tests a new blood pressure drug on 500 patients and finds their average blood pressure dropped by 12 points. She concludes the drug will lower blood pressure for the general population. Identify the sample, population, statistic, and parameter in this scenario. Which part is descriptive, and which is inferential?
2. **Data Classification:** Classify the following variables as Categorical (Nominal/Ordinal) or Numerical (Discrete/Continuous):
 - (a) The number of emails you receive in a day.
 - (b) The letter grades (A, B, C, D, F) of a class.
 - (c) The exact volume of water in a bottle (in ml).
 - (d) The brand of laptop you use (Apple, Dell, Lenovo).
3. **Scales of Measurement:** You are analyzing a dataset of used cars. Identify the scale of measurement (Nominal, Ordinal, Interval, Ratio) for each column:
 - (a) `Manufacturing_Year` (e.g., 2015, 2018)
 - (b) `Mileage_Driven` (e.g., 45000 km, 0 km)
 - (c) `Color` (e.g., Red, Silver)
 - (d) `Condition_Rating` (1 = Poor, 2 = Fair, 3 = Excellent)
4. **Data Structures:** A hospital records the daily body temperature of Patient X for a week. Meanwhile, a different researcher records the body temperatures of 100 different patients entering the ER on a Monday morning. Which dataset is Time Series, and which is Cross-Sectional?
5. **Data Science Application:** You are building a machine learning model to predict house prices. You have a feature called `Pincode` (e.g., 600036). Even though it is made of numbers, explain why treating it as a Numerical Continuous variable on a Ratio scale is a catastrophic statistical error.

Chapter 2

Describing Categorical Data

This chapter focuses on organizing and visualizing categorical variables—variables whose values are labels or group memberships rather than continuous numbers. In data science, categorical data is extremely common (e.g., predicting if an email is "Spam" or "Not Spam", classifying images as "Dog" or "Cat").

2.1 Frequency Distribution Tables

To make sense of raw categorical data, we must first summarize it into counts.

Definition: Frequency, Relative Frequency, and Percentage

Given a categorical variable observed n times:

- **Frequency (f_i):** The absolute count of observations falling into category i .
- **Relative Frequency (rf_i):** The proportion of observations falling into category i :

$$rf_i = \frac{f_i}{n}$$

- **Percentage:** Simply $rf_i \times 100\%$.
- **Mathematical Axiom:** The sum of all frequencies equals the total sample size n . The sum of all relative frequencies must exactly equal 1.

Data Science Relevance: In classification algorithms, checking the relative frequency of your target categories is the most important first step. If you have a dataset with 99% "Not Spam" and 1% "Spam", your dataset suffers from **Class Imbalance**. An algorithm could achieve 99% accuracy simply by guessing "Not Spam" every time, without actually learning anything!

Video Resource: IITM BS Lecture: W2_L1 Frequency Distributions



Official lecture detailing how to build frequency and relative frequency tables from raw categorical data.

Example: Frequency Table for a Categorical Variable

50 students chose their favourite subject: Maths (18), Stats (14), CS (12), English (6). Let's build a frequency table.

Subject	Frequency	Relative Freq.	Percentage
Maths	18	0.36	36%
Stats	14	0.28	28%
CS	12	0.24	24%
English	6	0.12	12%
Total	50	1.00	100%

```

1 import pandas as pd
2
3 # Create a Series of categorical data
4 subjects = pd.Series(['Maths']*18 + ['Stats']*14 + ['CS']*12 + ['English']*6)
5
6 # Calculate Absolute Frequencies
7 freq_table = subjects.value_counts()
8 print(freq_table)
9
10 # Calculate Relative Frequencies (Proportions)
11 rel_freq_table = subjects.value_counts(normalize=True)
12 print(rel_freq_table)

```

Listing 2.1: Calculating Frequencies in Pandas

2.2 Charts for Categorical Data

Visualizing categorical data helps humans instantly grasp the distribution.

Theorem/Rule: Choosing the Right Chart

- **Bar Chart:** Displays frequency or relative frequency on the y-axis; each category gets a separate bar. There are always gaps between the bars to indicate the categories are distinct. Best for comparing magnitudes across categories.
- **Pie Chart:** Shows each category as a sector proportional to its relative frequency. Best for showing part-to-whole relationships. Use sparingly human perception of exact angles is notoriously poor compared to perceiving heights.
- **Pareto Chart:** A specialized bar chart sorted in descending order of frequency, with a cumulative frequency curve overlaid. Useful for identifying the "vital few" categories that cause the most impact (e.g., the 80/20 rule in defect analysis).

Video Resource: IITM BS Lecture: W2_L2 Charts of Categorical Data 

Official lecture demonstrating how to properly construct Bar Charts and Pie Charts.

2.3 Mode and Median for Categorical Variables

How do we find the "average" of categorical data? We cannot mathematically sum "Maths" + "English" and divide by two. Instead, we rely on the Mode and Median.

Definition: Mode and Median for Categorical Data

- **Mode:** The category that appears most frequently (has the highest f_i). It is always defined for categorical data. A dataset can be *unimodal* (one mode), *bimodal*, or *multi-modal*.
- **Median for Ordinal Data:** The middle category when observations are arranged in their natural, logical order. Because ordinal data has a meaningful ordering (e.g., Small, Medium, Large), a median is mathematically defined. **The median is NEVER defined for nominal data.**

Video Resource: IITM BS Lecture: W2_L5 Mode & Median



Official lecture on calculating central tendencies for Nominal vs. Ordinal categorical variables.

Core Intuition: The Robustness of the Median

The Median is often preferred over the Mean (average) because it is highly **robust to outliers**. Imagine 5 people in a room making \$40k, \$50k, \$60k, \$70k, and \$80k. Both the median and mean are \$60k.

Now, Elon Musk walks in (making \$20 Billion). The mean skyrockets to over \$3 Billion, completely misrepresenting the typical person in the room! But the median simply shifts from the 3rd person (\$60k) to between the 3rd and 4th person (\$65k). The median doesn't care *how* extreme the edges are, it only cares about the physical order of the data.

Example: Mode and Median of Ordinal Data

40 customer satisfaction ratings on the scale Poor/Fair/Good/Excellent were recorded: Poor (5), Fair (8), Good (18), Excellent (9).

- **Mode:** Good (highest frequency = 18).
- **Median:** First, arrange the categories in order. Total $n = 40$, so the median lies between the 20th and 21st observations. Cumulative counts: Poor (5), Fair (5+8=13), Good (13+18=31). Because both the 20th and 21st values fall within the "Good" category (which spans from the 14th to 31st observation), the Median = **Good**.

2.4 Best Practices for Graphing

Data visualizations can be highly misleading if not constructed ethically and correctly.

Theorem/Rule: Graph Design Principles

- Always label both axes with variable names and units.
- **Start the y-axis at zero** for bar charts. Truncating the y-axis exaggerates tiny differences between categories, leading to misleading visual comparisons.
- Use consistent color schemes and avoid using color *alone* to convey information (for accessibility/color-blindness).
- Avoid 3-D effects and unnecessary "chart junk" (excessive gridlines, borders, shadows) that obscure the data.

Supplementary Lecture Videos

- [W2_L3: Describing categorical data - best practices while graphing data - 1](#)
- [W2_L4: Describing categorical data - best practices while graphing data - 2](#)

2.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Frequency Calculations:** In a survey of 200 people regarding their preferred smartphone OS, 110 chose Android, 80 chose iOS, and the rest chose Other. Calculate the relative frequency and percentage for each category.
2. **Charting Logic:** You are tasked with presenting the market share of 5 different car companies. You want your audience to quickly grasp that Company A holds exactly half (50%) of the total market. Which chart (Bar Chart or Pie Chart) is better suited to convey this specific message? Why?
3. **Graphing Ethics:** A company releases a bar chart comparing its sales to a competitor's. The competitor's bar has a height of 10,000, and the company's bar has a height of 11,000. However, the company's bar looks visually twice as tall as the competitor's bar. What cardinal rule of graphing did they violate?
4. **Mode and Median:** A t-shirt store sells 50 shirts in the following sizes: Small (10), Medium (20), Large (15), Extra Large (5).
 - (a) What is the mode?
 - (b) What is the median?
5. **Data Science Application:** You are classifying tumor images as Benign or Malignant. In your dataset of 10,000 images, 9,800 are Benign and 200 are Malignant. You write a terrible AI model that simply outputs "Benign" for every single image it sees, without even analyzing the picture. What is the accuracy of your model? What is this dataset problem called?

Chapter 3

Describing Numerical Data

Numerical data requires a richer mathematical toolkit than categorical data. Because the data consists of actual numbers on an interval or ratio scale, we can perform arithmetic. When analyzing numerical data, we generally want to capture three things: the *centre* (where is the middle?), the *spread* (how scattered is it?), and the *shape* (is it symmetric or skewed?).

3.1 Frequency Tables and Histograms

For discrete data with a massive range, or for continuous data (which has infinite possible values), we cannot simply list every unique value like we did for categorical data. Instead, we must group values into **class intervals** (or "bins").

Definition: Class Intervals and Histograms

- **Class Intervals:** Divide the total range of the data into k non-overlapping, equal-width bins (e.g., $0 - 10$, $10 - 20$, $20 - 30$).
- **Histogram:** A bar chart specifically designed for numerical data. The x-axis is a continuous number line. Crucially, the bars are *adjacent* (no gaps between them) to represent the continuous nature of the data.
- **Choosing Bins (k):** A common rule of thumb is $k \approx \sqrt{n}$, where n is the sample size. If you have 100 data points, use about 10 bins.

Video Resource: IITM BS Lecture: W3_L1 Frequency Tables for Numerical Data



Official lecture explaining how to bin continuous data and construct a histogram.

3.2 Measures of Central Tendency

Where is the "middle" of the data? We have three primary ways to define it.

Key Formula: Mean, Median, and Mode

Let $x_1, x_2, \dots, x_n$ be n numerical observations.

- **Sample Mean ($\bar{x}$):** The arithmetic average.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Median:** The literal middle value when observations are sorted in ascending order.
 - If n is odd: The exact middle value at position $\frac{n+1}{2}$.
 - If n is even: The average of the two middle values at $\frac{n}{2}$ and $\frac{n}{2} + 1$.
- **Mode:** The most frequently occurring value. (Less commonly used for continuous numerical data unless binned).

Theorem/Rule: Mean vs. Median: The Impact of Outliers

- The **mean** incorporates the exact value of *every single data point* into its calculation. Therefore, it is highly sensitive to **outliers** (extreme values).
- The **median** depends *only* on the rank-ordering of the data. Changing the largest value in the dataset from 100 to 1,000,000 does not change the median at all. Thus, the median is **robust** to outliers.
- **Right-Skewed (Long Right Tail):** Mean > Median. (The few huge values pull the mean up).
- **Left-Skewed (Long Left Tail):** Mean < Median. (The few tiny values pull the mean down).
- **Symmetric:** Mean $\approx$ Median.

Data Science Relevance: In data science, replacing missing values (NaNs) is called **Imputation**. If your feature has extreme outliers (like Income), imputing missing values with the *Mean* will severely distort your dataset. You must impute using the *Median*.

Video Resource: IITM BS Lecture: W3_L2 & L3 Mean, Median, and Mode



Official lectures comparing the formulas and structural differences between Mean and Median.

3.3 Measures of Dispersion (Spread)

Knowing the centre isn't enough. A city where the temperature is exactly 25°C every day has a mean of 25. A city where the temperature violently swings between 0°C and 50°C also has a mean of 25. We need to measure the spread.

Key Formula: Range, Variance, Standard Deviation, and IQR

- **Range:** $R = \max - \min$. Simplest measure, but entirely dictated by the two most extreme outliers.
- **Sample Variance (s^2):** The average squared distance of each point from the mean.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Note: We divide by $n - 1$ (Bessel's correction) rather than n to make it an unbiased estimator of the population variance.

- **Standard Deviation (s):** $s = \sqrt{s^2}$. This returns the spread back to the original units of the data.
- **Interquartile Range (IQR):**

$$\text{IQR} = Q_3 - Q_1$$

This measures the exact spread of the *middle 50%* of the data. Because it ignores the top 25% and bottom 25%, it is highly **robust** to outliers.

Data Science Relevance: Machine Learning algorithms like K-Means Clustering and Support Vector Machines rely on measuring distances between data points. If one feature has a standard deviation of 1000 and another has a standard deviation of 1, the algorithm will completely ignore the smaller feature. We use the Standard Deviation to mathematically scale (Standardize) all features to have $s = 1$.

Video Resource: IITM BS Lecture: W3_L4 Measures of Dispersion



Official lecture deriving the Variance and explaining why we square the distances from the mean.

3.4 Quartiles and the Five-Number Summary

We can slice ordered data into percentages to understand its structure deeply.

Definition: Percentiles and Quartiles

- **p -th Percentile:** The value below which $p\%$ of the observations fall.
- **Quartiles:** Specific percentiles that divide the sorted data into four equal quarters (25% each).
 - Q_1 (25th percentile) Lower quartile.
 - Q_2 (50th percentile) Median.
 - Q_3 (75th percentile) Upper quartile.
- **Five-Number Summary:** $\{\min, Q_1, Q_2, Q_3, \max\}$.
- **Boxplot:** A visual display of the Five-Number Summary. The "box" goes from Q_1 to Q_3 , containing the middle 50% of data.

Video Resource: IITM BS Lecture: W3_L5 Percentiles, Quartiles, and IQR 

Official lecture on calculating Q_1 and Q_3 and using them to mathematically flag Outliers.

3.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Central Tendency:** A startup has 5 employees. Four engineers make Rs. 80,000 each. The CEO makes Rs. 1,680,000.
 - (a) Calculate the Mean salary and the Median salary.
 - (b) The CEO goes to the press and says, "Our average employee makes Rs. 400,000! We pay great!" Explain mathematically why this is highly misleading, and state whether the distribution is Left-Skewed or Right-Skewed.
2. **Calculating Spread:** Given the dataset $X = \{2, 4, 4, 4, 5, 5, 7, 9\}$ ($n = 8$):
 - (a) Calculate the Sample Mean ($\bar{x}$).
 - (b) Calculate the Sample Variance (s^2) and Standard Deviation (s).
3. **Quartiles and IQR:** Using the same sorted dataset $X = \{2, 4, 4, 4, 5, 5, 7, 9\}$:
 - (a) Find the Median (Q_2).
 - (b) Find Q_1 and Q_3 . (Hint: Q_1 is the median of the lower half, Q_3 is the median of the upper half).
 - (c) Calculate the Interquartile Range (IQR).
4. **Data Science Application:** You are preprocessing a dataset of house prices for a neural network. You notice that the "Number of Bedrooms" feature ranges from 1 to 5, but the "Square Footage" feature ranges from 800 to 12,000. Which measure of dispersion dictates that you MUST apply feature scaling before training the model? What would happen if you didn't?
5. **Histograms vs. Bar Charts:** Explain the fundamental visual difference between a Histogram and a Bar Chart, and explain why that visual difference exists based on the underlying data types.

Chapter 4

Association Between Two Variables

Real-world data science problems almost always involve studying *relationships* between variables. Does smoking cause lung cancer? Does square footage predict house price? This chapter covers the mathematical tools to quantify association between two variables, depending on whether they are categorical or numerical.

4.1 Two Categorical Variables: Contingency Tables

When both variables are categorical, we use a **contingency table** (also known as a cross-tabulation or two-way table) to map their joint frequencies.

Definition: Contingency Table Frequencies

Let f_{ij} be the frequency in row i and column j , and let n be the total sample size.

- **Joint Relative Frequency:** $\frac{f_{ij}}{n}$ The proportion of the total sample that falls into both row i AND column j .
- **Marginal Relative Frequency:** The row or column totals divided by n . This effectively ignores one variable to just look at the other.
- **Conditional Relative Frequency:** The frequency of one variable *given* a specific category of the other. E.g., $\frac{f_{ij}}{f_{i\bullet}}$ (where $f_{i\bullet}$ is the row i total) gives the distribution of the column variable restricted only to row i .

Theorem/Rule: Defining Association for Categorical Variables

Two categorical variables are **associated** (dependent) if the conditional relative frequencies of one variable *change* depending on the category of the other variable. If the conditional distributions are identical across all rows, the variables are **independent**.

Video Resource: IITM BS Lecture: W4_L2 & L3 Two Categorical Variables 

Official lectures covering contingency tables and calculating relative frequencies.

Example: Contingency Table Analysis

200 people were surveyed on Gender (Male/Female) and Product Preference (A/B/C):

	Product A	Product B	Product C	Total
Male	40	30	50	120
Female	20	40	20	80
Total	60	70	70	200

Check for Association:

- Conditional distribution of preference *given Male*: $A = 40/120 \approx 33\%$, $B = 25\%$, $C = 42\%$.
- Conditional distribution of preference *given Female*: $A = 20/80 = 25\%$, $B = 50\%$, $C = 25\%$.

Because $33\% \neq 25\%$ (for Product A), the conditional distributions differ. Therefore, Gender and Product Preference are **associated**.

4.2 Two Numerical Variables: Covariance and Correlation

When both variables are numerical, we first visualize them using a **scatterplot**. Then, we quantify their linear relationship using Covariance and Correlation.

Key Formula: Covariance

The **covariance** between X and Y measures how much they vary together:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- $s_{XY} > 0$: X and Y tend to increase together (positive association).
- $s_{XY} < 0$: When X increases, Y tends to decrease (negative association).
- $s_{XY} = 0$: No linear association.

Fatal Flaw: The magnitude of s_{XY} depends entirely on the units of measurement. Covariance in grams will be mathematically smaller than covariance in kilograms, even though the relationship is identical.

Video Resource: IITM BS Lecture: W4_L6 Covariance



Official lecture deriving the Covariance formula and showing its unit-dependence flaw.

To fix the unit-dependence flaw of Covariance, we divide it by the standard deviations of X and Y to create a unitless metric.

Key Formula: Pearson Correlation Coefficient (r)

$$r = \frac{s_{XY}}{s_X \cdot s_Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

- $-1 \leq r \leq 1$ always.
- $r = +1$: Perfect positive linear relationship (all points lie exactly on an upward line).
- $r = -1$: Perfect negative linear relationship.
- $r = 0$: No **linear** relationship. (Warning: they could still have a perfect non-linear relationship, like a U-shape parabola!)

Data Science Relevance: Correlation is the foundation of **Feature Selection**. If a feature has $r \approx 0$ with your target variable, it's generally useless for a linear regression model and should be dropped. Conversely, if two input features have $r \approx 1$ with each other, they are redundant. This is called **Multicollinearity**, and you should drop one of them to simplify your model.

Video Resource: IITM BS Lecture: W4_L7 Correlation 

Official lecture on calculating Pearson's r and understanding bounds of -1 to 1 .

4.3 Categorical & Numerical: Point Biserial Correlation

What if one variable is Categorical (specifically, binary) and the other is Numerical? We encode the binary variable as 0 and 1, and apply the exact same Pearson formula. Algebraically, this simplifies into the **Point Biserial Correlation** (r_{pb}):

$$r_{pb} = \frac{\bar{x}_1 - \bar{x}_0}{s_X} \sqrt{\frac{n_1 n_0}{n^2}}$$

Where $\bar{x}_1, \bar{x}_0$ are the means of the continuous variable for group 1 and group 0, n_1, n_0 are the group sizes, and s_X is the overall standard deviation of the continuous variable.

Supplementary Lecture Videos

- [W4_L4: Association between two numerical variables - scatterplot](#)
- [W4_L8: Association between two numerical variables - fitting a line](#)
- [W4_L9: Association between categorical & numerical variables](#)

4.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Contingency Tables:** A university surveys 300 students on their housing (On-Campus vs. Off-Campus) and commuting method (Walk, Bike, Car).
 - Walk: 80 On-Campus, 20 Off-Campus
 - Bike: 20 On-Campus, 40 Off-Campus
 - Car: 10 On-Campus, 130 Off-Campus
 - (a) Construct the contingency table with marginal totals.
 - (b) Calculate the conditional relative frequency of commuting method *given* the student lives On-Campus.
 - (c) Are housing and commuting method associated? Mathematically justify your answer.
2. **Correlation Calculation:** Calculate the Pearson correlation coefficient (r) for the following dataset of hours studied (X) and test scores (Y):

$$X = \{1, 2, 3, 4, 5\}, \quad Y = \{2, 4, 5, 4, 5\}$$

(Hint: $\bar{x} = 3, \bar{y} = 4$)

3. **Correlation Fallacies:** A researcher plots the relationship between Ice Cream Sales (X) and Shark Attacks (Y). She calculates $r = +0.85$.
 - (a) Interpret the strength and direction of this linear relationship.
 - (b) The researcher concludes that eating ice cream causes sharks to attack. Identify the statistical fallacy she committed and provide the real logical explanation (the "confounding variable").
4. **Data Science Application:** You are training a Multiple Linear Regression model to predict a used car's price. You have two features: `Mileage_in_Miles` and `Mileage_in_Kilometers`.
 - (a) What is the exact Pearson correlation (r) between these two features?
 - (b) Should you include both features in your machine learning model? What is this scenario called?
5. **Non-Linear Relationships:** Plot the points $X = \{-2, -1, 0, 1, 2\}$ and $Y = \{4, 1, 0, 1, 4\}$.
 - (a) What is the mathematical relationship between X and Y ?
 - (b) Without doing the full math, what is the Pearson correlation coefficient (r) for this dataset? Why?

Chapter 5

Permutations and Combinations

Before we can compute probabilities, we must master the art of counting. In Data Science, if you want to know the probability of an event occurring by random chance, you must mathematically count the total number of possible scenarios. This chapter develops the foundational counting tools: the Addition/Multiplication principles, Factorials, Permutations, and Combinations.

5.1 Fundamental Counting Principles

At the heart of all complex counting are two simple, logical rules.

Theorem/Rule: Addition and Multiplication Principles

- **Addition Principle (OR Rule):** If Task A can be done in m ways, and Task B can be done in n ways, and the two tasks *cannot* be done simultaneously (they are mutually exclusive), then doing either Task A **OR** Task B can be done in $m + n$ ways.
- **Multiplication Principle (AND Rule):** If Task A can be done in m ways, and for *each* of those ways, a subsequent Task B can be done in n ways, then doing both Task A **AND** Task B together can be done in $m \times n$ ways.

This cascades across k consecutive tasks:

$$\text{Total ways} = n_1 \times n_2 \times \cdots \times n_k$$

Data Science Relevance: Machine Learning engineers use the Multiplication Principle constantly when performing **Grid Search Hyperparameter Tuning**. If you want to test an algorithm across 3 Learning Rates, 4 Batch Sizes, and 2 Optimizer types, the Multiplication Principle dictates you must train exactly $3 \times 4 \times 2 = 24$ distinct models to test every combination.

Video Resource: IITM BS Lecture: W5_L1 Basic Principles of Counting



Official lecture explaining the Addition and Multiplication principles with everyday examples.

5.2 Factorials

The Multiplication Principle naturally leads to factorials when we want to arrange a set of objects where we use one up at each step.

Definition: Factorial ($n!$)

For a non-negative integer n , the **factorial** $n!$ represents the number of ways to arrange n distinct objects in a sequence.

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1$$

Special Rule: $0! = 1$ (There is exactly one way to arrange zero objects: doing nothing).

Video Resource: IITM BS Lecture: W5_L2 Factorials 

Official lecture on calculating and simplifying ratios of factorials.

5.3 Permutations (Order Matters)

A permutation is an arrangement of a subset of items where the **order matters**. (e.g., selecting a President, Vice President, and Secretary from a 10-person committee).

Key Formula: Permutations of Distinct Objects (${}^n P_r$)

The number of ways to choose and arrange r distinct objects from a pool of n distinct objects is:

$${}^n P_r = \frac{n!}{(n - r)!}$$

Why? You start with $n!$ (total arrangements), but divide out $(n - r)!$ to mathematically erase the arrangements of the items you didn't pick.

Video Resource: IITM BS Lecture: W5_L3 Permutations (Distinct) 

Official lecture on ${}^n P_r$ and calculating ordered subsets.

Theorem/Rule: Permutations with Identical Objects

If you want to arrange n total objects, but some objects are identical indistinguishable copies of each other (e.g., arranging the letters in the word "MISSISSIPPI"), the formula is:

$$\frac{n!}{n_1! \times n_2! \times \cdots \times n_k!}$$

where n_i is the count of each identical subset. We divide by $n_i!$ because swapping identical letters doesn't create a "new" arrangement.

Video Resource: IITM BS Lecture: W5_L4 Permutations (Not Distinct) 

Official lecture detailing the exact math for arranging words with repeating letters.

5.4 Combinations (Order Doesn't Matter)

A combination is a selection of items where the **order does NOT matter**. (e.g., selecting a 3-person cleaning crew from a 10-person committee; picking Person A, B, and C is the exact same crew as picking B, C, and A).

Key Formula: Combinations (nC_r or $\binom{n}{r}$)

The number of ways to choose r objects from a pool of n objects, regardless of order, is:

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Why? It is exactly the Permutation formula nP_r , but we additionally divide by $r!$ to erase the internal ordering of the r chosen items, collapsing all their permutations into a single "Combination".

Video Resource: IITM BS Lecture: W5_L5 Combinations 

Official lecture explaining nC_r and the mathematical relationship between Permutations and Combinations.

Supplementary Lecture Videos

- [W5_L6: Permutations & combinations - applications](#)

5.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Basic Counting:** A restaurant offers a "Build a Burger" menu. You must choose exactly 1 type of bun (out of 3 options), exactly 1 type of meat (out of 4 options), and either choose to add a side of fries OR a side of onion rings (but not both). How many distinct burger meals can you build?
2. **Permutations (Distinct):** A track race has 8 runners.
 - (a) How many different ways can the runners finish the race in 1st, 2nd, and 3rd place? (Assuming no ties).
 - (b) Which formula did you use and why?
3. **Permutations (Non-Distinct):** You are trying to train a text generation model. To test it, you ask it to generate every unique anagram of the word **STATISTICS**.
 - (a) How many unique arrangements of the word exist?
 - (b) Explain mathematically why dividing by the factorials in the denominator is necessary.
4. **Combinations:** You are building an ensemble Machine Learning model. You have trained 10 different baseline algorithms (Decision Tree, SVM, Logistic Regression, etc.). You want to randomly select exactly 4 of them to vote together in your ensemble.
 - (a) How many different combinations of 4 models can you create?
 - (b) If you decided to pick 6 models instead of 4, how many combinations would exist? Why is this number identical to the previous answer?
5. **Advanced Application:** A committee of 5 people is to be formed from a group of 6 men and 7 women.
 - (a) How many committees can be formed if there are no restrictions?
 - (b) How many committees can be formed that contain exactly 3 men and 2 women?

Chapter 6

Probability Basics

With a firm grasp of counting principles, we now cross over from descriptive statistics into the realm of uncertainty: Probability. In data science, probability is the mathematical language we use to quantify confidence in our model's predictions.

6.1 Random Experiments and Sample Space

Definition: Random Experiment and Sample Space

- **Random Experiment:** Any process or action whose exact outcome cannot be predicted with certainty, but whose set of all possible outcomes is known. (Example: Flipping a coin, rolling a die, a user clicking on an ad).
- **Sample Space (S or Ω):** The set of *all* possible outcomes of a random experiment.
- **Sample Point (ω):** A single, specific outcome within the sample space.

Video Resource: IITM BS Lecture: W6_L1 & L3 Basic Definitions & Sample Space 

Official lectures defining random experiments, outcomes, and building the Sample Space S .

6.2 Events and Set Operations

Definition: Events

An **Event (A)** is any subset of the sample space ($A \subseteq S$). We say the event A *occurred* if the final outcome of the experiment is an element of A .

- **Simple Event:** An event consisting of exactly one outcome.
- **Compound Event:** An event consisting of more than one outcome.
- **Impossible Event ($\emptyset$):** The empty set. An event that contains no outcomes.
- **Certain Event (S):** The entire sample space. It contains all outcomes and must occur.

Because events are sets, we manipulate them using Set Theory operations:

Theorem/Rule: Set Operations on Events

- **Union ($A \cup B$):** The event that A occurs, OR B occurs, OR both occur.
- **Intersection ($A \cap B$):** The event that BOTH A and B occur simultaneously.
- **Complement (A^c or A'):** The event that A does *not* occur. ($A^c = S - A$).
- **Mutually Exclusive (Disjoint):** Two events that cannot happen at the same time. Mathematically, $A \cap B = \emptyset$.

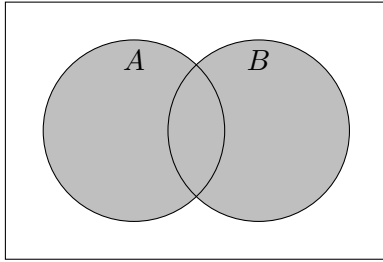


Figure 6.1: *
Union ($A \cup B$)

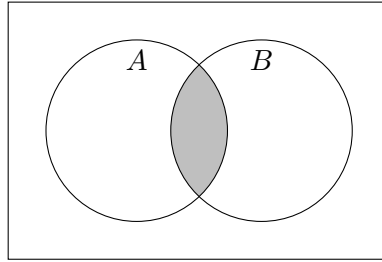


Figure 6.2: *
Intersection ($A \cap B$)

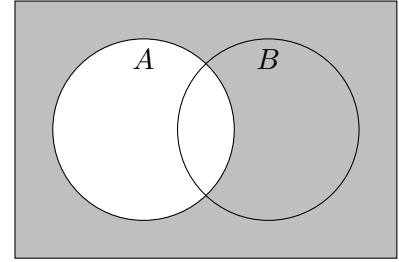


Figure 6.3: *
Complement (A^c)

Video Resource: IITM BS Lecture: W6_L2 Events & Basic Operations



Official lecture explaining how to map English language logic (AND, OR, NOT) into Set Theory operations ($\cap, \cup, ^c$).

6.3 Axioms and Properties of Probability

Probability is a function that assigns a real number $P(A)$ to every event A , quantifying how likely it is to occur. To be a mathematically valid probability, the function must satisfy Kolmogorov's Axioms.

Key Formula: The Three Axioms of Probability

1. **Non-negativity:** For any event A , the probability is at least zero:

$$P(A) \geq 0$$

2. **Normalization:** The probability of the entire sample space is exactly one:

$$P(S) = 1$$

3. **Additivity:** If A and B are mutually exclusive ($A \cap B = \emptyset$), then the probability of their union is the sum of their probabilities:

$$P(A \cup B) = P(A) + P(B)$$

Core Intuition: Probability as a Mass

To intuitively understand Kolmogorov's Axioms, think of probability as **physical mass**. You are handed exactly 1.0 kg of modeling clay (the total probability, Axiom 2). You must distribute this clay across all the possible outcomes in your sample space.

You can give an outcome zero clay, or some positive amount, but you cannot give an outcome "negative" clay (Axiom 1). If you want to know the weight of a combined event (like "rolling an even number"), you simply put the 2-outcome, 4-outcome, and 6-outcome on a scale and add their weights together (Axiom 3).

From these three fundamental axioms, we can logically derive several critical properties used constantly in data science.

Theorem/Rule: Derived Properties of Probability

- **Probability Bounds:** $0 \leq P(A) \leq 1$.
- **Complement Rule:** $P(A^c) = 1 - P(A)$. (Extremely useful when calculating $P(A)$ directly is too difficult, but calculating the probability of it *not* happening is easy).
- **General Addition Rule (Inclusion-Exclusion):** For *any* two events A and B (even if they are not mutually exclusive):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

We subtract the intersection so we don't double-count the outcomes where both occur!

Data Science Relevance: Machine Learning classifiers (like Logistic Regression) output predictions as probabilities between 0 and 1. The **Softmax function** in Neural Networks specifically forces the output layer to obey Kolmogorov's Second Axiom: the sum of the probabilities for all possible predicted classes (the Sample Space S) must equal exactly 1.0.

Video Resource: IITM BS Lecture: W6_L4 Properties of Probability



Official lecture deriving the Complement Rule and the General Addition Rule from the three axioms.

6.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Sample Spaces:** Define the sample space S for the following random experiments:
 - (a) Flipping a coin three times.
 - (b) Counting the number of cars that pass through a toll booth in an hour.
 - (c) Measuring the exact lifespan of a lightbulb (in hours).
2. **Set Operations:** You roll a standard 6-sided die. Let $S = \{1, 2, 3, 4, 5, 6\}$. Let Event A be rolling an even number. Let Event B be rolling a number greater than 3.
 - (a) Define the sets for A and B .
 - (b) Find $A \cap B$ and describe it in plain English.
 - (c) Find $A \cup B$.
 - (d) Find A^c .
 - (e) Are A and B mutually exclusive? Why or why not?
3. **Complement Rule:** A cybersecurity system is monitoring 5 independent incoming network packets. The probability that a specific packet gets dropped by the firewall is 0.1. You want to find the probability that **at least one** packet gets dropped.
 - (a) Why is calculating this directly using the Addition Rule extremely tedious?
 - (b) Use the Complement Rule to solve this elegantly. (Hint: What is the complement of "at least one packet drops"?)
4. **General Addition Rule:** In a class of 100 students, 60 study Python (P), 50 study R (R), and 20 study both.
 - (a) What is the probability that a randomly selected student studies Python OR R?
 - (b) What is the probability that a student studies neither?
5. **Axioms Check:** A machine learning model outputs the following probabilities for classifying an image of an animal:

$$P(\text{Dog}) = 0.5, \quad P(\text{Cat}) = 0.4, \quad P(\text{Bird}) = 0.2$$

Assuming these are the only three possible classes, which of Kolmogorov's Axioms does this model violate? Explain why.

Chapter 7

Conditional Probability and Bayes' Theorem

In the real world, probabilities do not exist in a vacuum. The probability of an event drastically changes when you receive new information. This chapter covers how to update probabilities based on new evidence, a concept that forms the absolute core of modern Machine Learning (Bayesian Inference).

7.1 Conditional Probability

Definition: Conditional Probability

The **conditional probability** of event A occurring *given* that event B has already occurred is denoted as $P(A | B)$. It is calculated by restricting the sample space to only the outcomes where B occurs:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0$$

Data Science Relevance: In natural language processing, predictive text uses conditional probability. What is the probability the next word is "Learning", *given* that the previous word was "Machine"? $P(\text{Learning} | \text{Machine})$.

Video Resource: IITM BS Lecture: W7_L1 & L2 Conditional Probability 

Official lectures defining conditional probability using contingency tables and deriving the formula.

7.2 The Multiplication Rule

By rearranging the conditional probability formula, we get the Multiplication Rule, which tells us how to find the probability of both events occurring simultaneously (their intersection).

Key Formula: The Multiplication Rule

$$P(A \cap B) = P(B) \times P(A | B)$$

Alternatively, because $A \cap B = B \cap A$:

$$P(A \cap B) = P(A) \times P(B | A)$$

Video Resource: IITM BS Lecture: W7_L3 Multiplication Rule 

Official lecture showing how to cascade the multiplication rule across multiple dependent events (like drawing cards without replacement).

7.3 Independent Events

What if knowing that event B occurred gives you absolutely zero information about whether event A will occur? (e.g., flipping a coin, then rolling a die).

Theorem/Rule: Mathematical Independence

Two events A and B are **independent** if and only if:

$$P(A | B) = P(A)$$

Substituting this into the Multiplication Rule gives the ultimate test for independence:

$$P(A \cap B) = P(A) \times P(B)$$

If this equation holds true, the events are independent. If it does not, they are dependent.

Data Science Relevance: The popular ML algorithm **Naive Bayes** assumes that all input features are strictly independent of each other given the target class. This assumption is "naive" (often false in real life), but makes the algorithm incredibly fast and computationally cheap.

Video Resource: IITM BS Lecture: W7_L4 & L6 Independent Events 

Official lectures covering the mathematical definition and properties of independent events.

7.4 Law of Total Probability and Bayes' Theorem

Often, an event A can happen through several different, mutually exclusive paths (partitions of the sample space). The **Law of Total Probability** allows us to sum up the probabilities of these paths.

Key Formula: Law of Total Probability

If $B_1, B_2, \dots, B_k$ form a partition of the sample space (they are mutually exclusive and their union is the whole space), then for any event A :

$$P(A) = \sum_{i=1}^k P(A \cap B_i) = \sum_{i=1}^k P(B_i)P(A | B_i)$$

What if we know A occurred, and we want to work *backwards* to find the probability that it came from path B_i ? This reverse-engineering is the most important theorem in all of statistics.

Theorem/Rule: Bayes' Theorem

$$P(B_i | A) = \frac{P(B_i) \times P(A | B_i)}{P(A)}$$

Using the Law of Total Probability to expand the denominator:

$$P(B_i | A) = \frac{P(B_i) \times P(A | B_i)}{\sum_{j=1}^k P(B_j)P(A | B_j)}$$

Example: Bayes' Theorem in Python (Medical Diagnosis / Spam Filter)

We can programmatically evaluate Bayesian posterior updates using a clean Python function:

```
1 def bayes_update(prior, likelihood_true, likelihood_false):
2     """
3     Computes P(Event | Evidence) using Bayes' Theorem
4     """
5     p_evidence = (prior * likelihood_true) + ((1 - prior) * likelihood_false)
6     posterior = (prior * likelihood_true) / p_evidence
7     return posterior
8
9 # Example: Disease prevalence 1%, Sensitivity 90%, False Positive 5%
10 prior_disease = 0.01
11 p_pos_given_disease = 0.90
12 p_pos_given_healthy = 0.05
13
14 post_disease = bayes_update(prior_disease, p_pos_given_disease,
15                             p_pos_given_healthy)
16 print(f"P(Disease | Positive Test): {post_disease:.4f}") # Output: ~0.1538
17                                     (15.38%)
```

Video Resource: IITM BS Lecture: W7_L7 Bayes' Rule



Official lecture deriving Bayes' Theorem and applying it to real-world scenarios.

7.5 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

- 1. Conditional Probability:** A standard 52-card deck is shuffled. You draw one card.
 - (a) What is the probability it is a King (K)?
 - (b) Your friend peeks at the card and says, "It is a face card" (F). Given this new information, what is the new conditional probability $P(K \mid F)$? (A deck has 12 face cards: 4 Jacks, 4 Queens, 4 Kings).
- 2. Multiplication Rule:** A bag contains 5 Red marbles and 3 Blue marbles. You draw two marbles consecutively *without replacement*. What is the probability of drawing a Red marble first, AND a Blue marble second?
- 3. Checking for Independence:** In a sample space, $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cup B) = 0.7$.
 - (a) Use the General Addition Rule to find $P(A \cap B)$.
 - (b) Are events A and B mathematically independent? Prove it.
- 4. Bayes' Theorem (Medical Testing):** A rare disease affects 1% of the population ($P(D) = 0.01$). A test for the disease is 90% accurate for positive patients ($P(+ \mid D) = 0.90$), but has a 5% false positive rate for healthy patients ($P(+ \mid H) = 0.05$).
 - (a) Find the total probability that a random person tests positive, $P(+)$.
 - (b) If a random person tests positive, what is the actual probability they have the disease, $P(D \mid +)$?
- 5. Data Science Application:** You are building a spam filter. 20% of all incoming emails are spam. 80% of spam emails contain the word "Free". However, 10% of legitimate emails also contain the word "Free". An email arrives containing the word "Free". What is the exact probability that it is spam?

Chapter 8

Random Variables

Until now, our sample spaces have been arbitrary sets of outcomes, like {Heads, Tails} or {Spam, Legit}. But mathematics requires numbers. A **Random Variable** acts as a bridge, translating physical outcomes into numbers so we can do algebra and calculus on them.

8.1 Introduction to Random Variables

Definition: Random Variable

A **Random Variable** (X) is a function that maps every possible outcome in the sample space S to a real number.

$$X : S \rightarrow \mathbb{R}$$

For example, if we flip two coins, $S = \{HH, HT, TH, TT\}$. We can define the Random Variable X as "the number of Heads". Then: $X(HH) = 2$, $X(HT) = 1$, $X(TH) = 1$, $X(TT) = 0$.

Theorem/Rule: Discrete vs. Continuous Random Variables

- **Discrete Random Variable:** Can only take on a countable number of distinct values. Often associated with *counting* things (e.g., number of emails received, number of defective products, rolling a die).
- **Continuous Random Variable:** Can take on any value within an infinitely dense interval. Often associated with *measuring* things (e.g., exact height, time, temperature).

Data Science Relevance: In Machine Learning, classifying a label (like predicting whether a tumor is Malignant or Benign) is modeling a **Discrete Random Variable**. Predicting an exact numerical value (like predicting the future price of a stock) is modeling a **Continuous Random Variable**.

Video Resource: IITM BS Lecture: W8_L1 & L3 Random Variables



Official lectures defining random variables and clearly separating Discrete from Continuous.

8.2 The Probability Mass Function (PMF)

For discrete random variables, we want to know the probability of hitting a specific numerical value.

Definition: Probability Mass Function (PMF)

The **Probability Mass Function** of a discrete random variable X , denoted $f(x)$ or $P(X = x)$, gives the exact probability that X will take on the value x .

Key Formula: Properties of a Valid PMF

For $f(x)$ to be a valid mathematical PMF, it must satisfy two Kolmogorov-derived rules:

1. **Non-negativity:** $f(x) \geq 0$ for all possible values of x . (You cannot have a negative probability).
2. **Sum to One:** $\sum f(x) = 1$. (The sum of probabilities for all possible values of X must equal exactly 1).

Video Resource: IITM BS Lecture: W8_L4 & L5 Probability Mass Function



Official lectures detailing the properties of PMFs and how to graph them as stick plots.

8.3 The Cumulative Distribution Function (CDF)

Sometimes we don't just want $P(X = x)$. We want to know the probability that X is *less than or equal to* x . (e.g., What is the probability a server goes down in 5 days *or less*?).

Definition: Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function**, denoted $F(x)$, gives the accumulating probability up to and including x .

$$F(x) = P(X \leq x) = \sum_{k \leq x} P(X = k)$$

Theorem/Rule: Properties of a Valid CDF

1. **Bounds:** $0 \leq F(x) \leq 1$.
2. **Non-Decreasing:** If $x_1 < x_2$, then $F(x_1) \leq F(x_2)$. (It can only go up, never down).
3. **Limits:** As $x \rightarrow -\infty$, $F(x) \rightarrow 0$. As $x \rightarrow \infty$, $F(x) \rightarrow 1$.
4. For a discrete variable, the CDF graph looks like a **step function**, jumping up exactly at the values where the PMF has mass.

Video Resource: IITM BS Lecture: W8_L6 Cumulative Distribution Function



Official lecture explaining how to compute the CDF from the PMF and how to read the step function graph.

8.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Discrete vs. Continuous:** Classify the following random variables as Discrete or Continuous:
 - (a) The number of customer complaints received in a day.
 - (b) The exact time it takes for a webpage to load.
 - (c) The number of heads when flipping 100 coins.
 - (d) The volume of water in a water bottle.
2. **Validating a PMF:** A discrete random variable X can take the values $x \in \{1, 2, 3, 4\}$. Its PMF is defined as $P(X = x) = c \cdot x$.
 - (a) Find the value of the constant c that makes this a valid PMF.
 - (b) What is the exact probability $P(X = 3)$?
3. **Creating a PMF:** You roll two standard 6-sided dice. Let the Random Variable X be the *sum* of the two dice.
 - (a) What is the Sample Space of the underlying experiment, and what is its size?
 - (b) What is the set of all possible numerical values that X can take?
 - (c) Calculate $P(X = 7)$ and $P(X = 12)$.
4. **Building a CDF:** Let X have the following PMF:
$$P(X = 1) = 0.2, \quad P(X = 2) = 0.5, \quad P(X = 3) = 0.3$$
 - (a) Calculate the CDF values: $F(1), F(2), F(3)$.
 - (b) What is the value of the CDF at $F(1.5)$? (Think carefully about the definition of $X \leq 1.5$).
 - (c) Sketch (mentally or on paper) what the graph of this CDF looks like.
5. **Using the CDF:** You are given the CDF of a discrete random variable Y that only takes integer values: $F(1) = 0.1, \quad F(2) = 0.4, \quad F(3) = 0.9, \quad F(4) = 1.0$ Using *only* the CDF, find the exact PMF probability $P(Y = 3)$.

Chapter 9

Expectation and Variance of Random Variables

Once we have defined a Random Variable and its Probability Mass Function (PMF), we want to summarize its behavior. Just like we summarized physical data using Mean and Variance in Chapter 3, we can mathematically summarize the theoretical distributions of Random Variables using **Expectation** and **Variance**.

9.1 Expectation (Expected Value)

Definition: Expected Value (Mean) of a Random Variable

The **Expected Value**, denoted $E[X]$ or μ_X , is the long-run average value of repetitions of the experiment it represents. For a discrete random variable, it is the probability-weighted sum of all possible values.

$$E[X] = \sum_x x \cdot P(X = x)$$

Note: $E[X]$ is not necessarily a value that X can actually take! For example, the expected value of a standard die roll is 3.5.

Theorem/Rule: Properties of Expectation (Linearity)

Expectation is a strictly linear operator. For any constants a and b , and any random variables X and Y :

- **Constants:** $E[c] = c$
- **Scaling and Shifting:** $E[aX + b] = aE[X] + b$
- **Addition:** $E[X + Y] = E[X] + E[Y]$ (This holds true *even if* X and Y are dependent!)

Data Science Relevance: In Reinforcement Learning, an AI agent takes actions to maximize its "Expected Return". Because the environment is stochastic (random), the AI cannot predict exactly what reward it will get, so it calculates the mathematical Expected Value of all possible future states and chooses the path with the highest $E[X]$.

Video Resource: IITM BS Lecture: W9_L2 & L3 Expectation 

Official lectures defining $E[X]$ and proving the Linearity of Expectation.

9.2 Variance of a Random Variable

While Expectation tells us the theoretical center, we also need to know the theoretical spread.

Definition: Variance and Standard Deviation

The **Variance** of a random variable X , denoted $Var(X)$ or σ_X^2 , measures the expected squared distance from the mean.

$$Var(X) = E[(X - \mu)^2]$$

A mathematically equivalent (and computationally faster) formula is:

$$Var(X) = E[X^2] - (E[X])^2$$

The **Standard Deviation** is simply the square root of the variance: $\sigma_X = \sqrt{Var(X)}$.

Theorem/Rule: Properties of Variance (Non-Linearity)

Unlike Expectation, Variance is **NOT** linear. For any constants a and b :

- **Constants:** $Var(c) = 0$ (A constant has no spread, it never varies).
- **Shifting:** $Var(X + b) = Var(X)$ (Adding a flat number to everything shifts the center, but does not change the spread).
- **Scaling:** $Var(aX) = a^2 Var(X)$ (Multiplying by a constant scales the variance by the *square* of that constant).
- **Addition:** $Var(X + Y) = Var(X) + Var(Y)$ **ONLY IF** X and Y are strictly independent.

Data Science Relevance: In Finance and Quantitative Trading, Expected Value represents your projected profit, and Variance represents your **Risk**. Modern Portfolio Theory aims to construct a basket of stocks that maximizes $E[X]$ while minimizing $Var(X)$.

Video Resource: IITM BS Lecture: W9_L4 & L5 Variance Properties 

Official lectures deriving $Var(X)$ and proving why $Var(aX) = a^2 Var(X)$.

9.3 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Calculating Expectation:** A lottery ticket costs Rs. 10 to play. You have a 1% chance of winning Rs. 500, a 5% chance of winning Rs. 50, and a 94% chance of winning nothing (Rs. 0). Let X be the gross payout of the ticket.
 - (a) What is the PMF of X ?
 - (b) Calculate the Expected Value $E[X]$.
 - (c) Factoring in the Rs. 10 cost to play, what is your expected net profit?
2. **Calculating Variance:** A discrete random variable Y has the following PMF: $P(Y = -1) = 0.2$, $P(Y = 0) = 0.5$, $P(Y = 1) = 0.3$.
 - (a) Calculate the Expected Value $\mu = E[Y]$.
 - (b) Calculate $E[Y^2]$.
 - (c) Calculate the Variance $Var(Y)$ using the computational formula.
3. **Properties of Expectation and Variance:** Let X be a random variable representing the temperature in Celsius. You know $E[X] = 25^\circ\text{C}$ and $Var(X) = 4$. You want to convert this to Fahrenheit using the formula $F = 1.8X + 32$.
 - (a) What is the Expected Value of the temperature in Fahrenheit, $E[F]$?
 - (b) What is the Variance of the temperature in Fahrenheit, $Var(F)$? (Be very careful with the properties).
 - (c) What is the Standard Deviation in Fahrenheit?
4. **Independence and Variance:** You flip a fair coin, where Heads = 1 and Tails = 0. Let this random variable be X . You roll a fair 6-sided die, yielding values 1 through 6. Let this random variable be Y .
 - (a) $E[X] = 0.5$ and $E[Y] = 3.5$. What is $E[X + Y]$?
 - (b) Are X and Y independent? What does this mean for $Var(X + Y)$?

Chapter 10

The Binomial Distribution

So far, we have looked at generic, custom-built Random Variables. However, many real-world problems follow the exact same mathematical structure. Instead of recalculating probabilities from scratch every time, statisticians have derived standard **Probability Distributions** that act as mathematical templates. The most fundamental of these is the Binomial Distribution.

10.1 The Bernoulli Distribution

Before we can build a Binomial Distribution, we must define its basic building block: a Bernoulli Trial.

Definition: Bernoulli Trial and Distribution

A **Bernoulli Trial** is a random experiment that has exactly two possible outcomes: "Success" (mapped to 1) and "Failure" (mapped to 0). Let the probability of success be p , and the probability of failure be $1 - p$ (often denoted q).

A Random Variable X follows a **Bernoulli Distribution**, denoted $X \sim \text{Bernoulli}(p)$, if its PMF is:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

Theorem/Rule: Expectation and Variance of Bernoulli

For $X \sim \text{Bernoulli}(p)$:

- **Expectation:** $E[X] = (1 \times p) + (0 \times (1 - p)) = p$
- **Variance:** $\text{Var}(X) = p(1 - p)$

Video Resource: IITM BS Lecture: W10_L1 Bernoulli Distribution



Official lecture defining the Bernoulli random variable and calculating its expected value and variance.

10.2 Independent and Identically Distributed (IID) Trials

A single coin flip is a Bernoulli trial. But what if we flip the coin 10 times? To mathematically chain these trials together, we must make two strict assumptions, known as **IID**.

Definition: IID (Independent and Identically Distributed)

A sequence of random variables $X_1, X_2, \dots, X_n$ is IID if:

1. **Independent:** The outcome of one trial has absolutely zero effect on the outcome of any other trial.
2. **Identically Distributed:** Every single trial has the exact same probability of success p . The coin does not suddenly become biased on the 5th flip.

Video Resource: IITM BS Lecture: W10_L2 IID Bernoulli Trials



Official lecture explaining the vital assumptions of Independence and Identical Distribution.

10.3 The Binomial Distribution

If we run n IID Bernoulli trials and count the *total* number of successes, that total count follows a **Binomial Distribution**.

Definition: Binomial Distribution

Let X be the total number of successes in n IID Bernoulli trials, each with success probability p . We write this as $X \sim \text{Bin}(n, p)$.

The Probability Mass Function (PMF) of X is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Core Intuition: Why the Combinations Term?

p^k gives the probability of k successes. $(1 - p)^{n-k}$ gives the probability of $n - k$ failures. But if we just multiply those, we are calculating the probability of a *specific sequence* (e.g. S-S-F-F).

But what if we flipped S-F-S-F? Or F-S-S-F? To find the probability of exactly k successes occurring in *any order*, we must multiply by the number of different ways to arrange k successes among n slots. That's exactly what the combinations formula $\binom{n}{k}$ does!

```
1 from scipy.stats import binom
2
3 # Parameters: n=10, p=0.5
4 # What is the exact probability of getting exactly 5 successes?
5 prob = binom.pmf(k=5, n=10, p=0.5)
6 print(f"P(X=5) = {prob:.4f}")
7
8 # What is the probability of getting AT MOST 5 successes? (CDF)
9 cum_prob = binom.cdf(k=5, n=10, p=0.5)
10 print(f"P(X<=5) = {cum_prob:.4f}")
```

Listing 10.1: Calculating Binomial Probabilities in SciPy

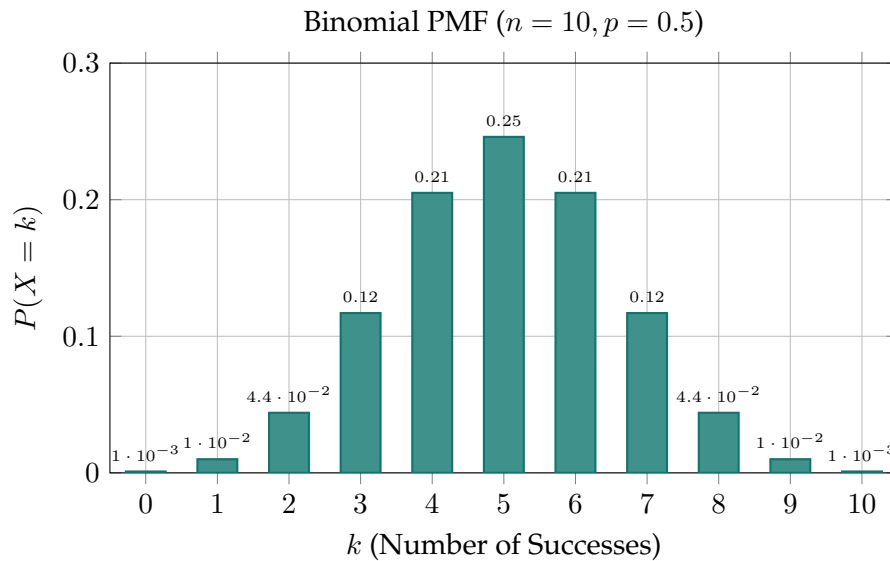


Figure 10.1: The beautiful symmetric bell-shape of the Binomial Distribution when $p = 0.5$.

Theorem/Rule: Expectation and Variance of Binomial

Because a Binomial random variable is simply the sum of n independent Bernoulli random variables ($X = X_1 + X_2 + \dots + X_n$), we can use the linearity of expectation and variance!

- **Expectation:** $E[X] = E[X_1] + \dots + E[X_n] = p + \dots + p = n \times p$
- **Variance:** $Var(X) = Var(X_1) + \dots + Var(X_n) = n \times p(1 - p)$

Data Science Relevance: The Binomial distribution is the foundation of **A/B Testing** and measuring **Conversion Rates**. If an e-commerce website has a conversion rate of $p = 0.05$, and $n = 1000$ users visit the site, the number of purchases made follows a Binomial Distribution. Furthermore, the most popular classification algorithm, **Logistic Regression**, is mathematically derived directly from the Bernoulli/Binomial distribution!

Video Resource: IITM BS Lecture: W10_L3, L4, L5 Binomial Distribution



Official lectures deriving the Binomial PMF, modeling real-world situations, and proving the Expectation/-Variance formulas.

10.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Bernoulli Trials:** An ad campaign has a Click-Through Rate (CTR) of 3%. Let X represent a single user viewing the ad, where $X = 1$ if they click and $X = 0$ if they don't.
 - (a) What is the distribution of X ?
 - (b) Calculate the Expected Value and Variance of X .
2. **Binomial PMF:** A factory produces lightbulbs. 10% of all lightbulbs produced are defective. You randomly sample 5 lightbulbs off the assembly line. Let X be the number of defective bulbs in your sample.
 - (a) What are the parameters n and p for this Binomial distribution?
 - (b) What is the exact probability of getting exactly 2 defective bulbs?
 - (c) What is the probability of getting exactly 0 defective bulbs?
3. **Binomial Expectation and Variance:** A multiple choice test has 50 questions. Each question has 4 options, with only 1 correct answer. A student completely guesses on every single question. Let Y be the number of questions they get correct.
 - (a) Calculate the expected number of correct answers, $E[Y]$.
 - (b) Calculate the variance of their score, $Var(Y)$.
4. **Using the Complement Rule with Binomial:** A medical treatment has a 60% success rate. It is administered to 10 patients.
 - (a) What is the probability that **at least one** patient is successfully cured? (Hint: Use the Complement rule!).
5. **Data Science Application:** You are running an A/B test on a website. You show a new checkout button to 100 users. You want to model the number of clicks using a Binomial distribution. What two **IID assumptions** must you mathematically make about the users for the Binomial model to be valid? Give an example of how one of those assumptions could be violated in real life.

Chapter 11

The Poisson Distribution

The Binomial distribution requires a fixed number of trials n . But what if we want to model events where n is effectively infinite and p is microscopically small? For example, how many emails will you receive in the next hour? The **Poisson Distribution** mathematically models the count of rare events occurring within a fixed interval of time or space.

11.1 The Poisson Limit of the Binomial

Imagine an e-commerce website with millions of daily visitors (n is huge). The probability that any single visitor buys a specific \$10,000 laptop is tiny (p is near zero). Calculating $\binom{n}{k} p^k (1-p)^{n-k}$ with $n = 1,000,000$ will crash most computers due to the massive factorials.

In 1837, Siméon Denis Poisson proved that if you take the mathematical limit of the Binomial distribution as $n \rightarrow \infty$ and $p \rightarrow 0$, while keeping the expected value $\lambda = np$ constant, the impossible factorials beautifully cancel out into an elegant exponential function.

Theorem/Rule: Poisson Approximation to the Binomial

If $X \sim \text{Bin}(n, p)$ with $n \geq 20$ and $p \leq 0.05$, you can safely approximate the probability using the Poisson distribution with parameter $\lambda = np$:

$$\text{Bin}(n, p) \approx \text{Pois}(\lambda = np)$$

11.2 The Poisson Distribution

Definition: Poisson Random Variable

Let X be the number of events occurring in a fixed interval. X follows a **Poisson Distribution**, denoted $X \sim \text{Pois}(\lambda)$, if the events occur with a known constant mean rate λ (lambda), and each event occurs independently of the time since the last event.

The Probability Mass Function (PMF) is:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad \text{for } k = 0, 1, 2, \dots$$

Where e is Euler's number (≈ 2.71828).

Theorem/Rule: Expectation and Variance of Poisson

The Poisson distribution has a very unique and mathematically elegant property: its expected value and its variance are identical.

- **Expectation:** $E[X] = \lambda$
- **Variance:** $Var(X) = \lambda$

Data Science Relevance: The Poisson distribution is heavily used in **Operations Research** and **Queueing Theory**. Amazon AWS uses Poisson processes to mathematically predict how many servers they need to spin up to handle incoming server requests per minute. Traffic engineers use it to model the number of cars arriving at a toll booth per hour.

11.3 Changing the Interval

The parameter λ is strictly tied to a specific interval length. If you change the length of the interval, you *must* scale λ linearly.

Example: Scaling Lambda

If a call center receives an average of $\lambda = 3$ calls per *minute*:

- What is the distribution for a *2-minute* interval?
The new rate is $\lambda_{\text{new}} = 3 \times 2 = 6$. So, $X \sim \text{Pois}(6)$.
- What is the distribution for a *20-second* interval?
20 seconds is $1/3$ of a minute. $\lambda_{\text{new}} = 3 \times (1/3) = 1$. So, $X \sim \text{Pois}(1)$.

Supplementary Video Resources

- [StatQuest: The Poisson Distribution, Clearly Explained](#)

11.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **Poisson Probabilities:** A bookstore sells an average of 2 copies of a rare textbook per week ($\lambda = 2$). Assuming sales follow a Poisson distribution:
 - (a) What is the exact probability that they sell exactly 3 copies in a given week?
 - (b) What is the probability that they sell exactly 0 copies?
2. **Scaling the Interval:** A server receives an average of 6 malicious DDoS requests per minute.
 - (a) What is the probability of receiving exactly 10 requests in a **2-minute** window? (Make sure to scale λ first!).
 - (b) What is the expected number of requests in a **10-second** window?
3. **Poisson Approximation to Binomial:** A machine produces computer chips, and 1% of them are defective ($p = 0.01$). You randomly sample 100 chips ($n = 100$).
 - (a) Calculate the exact Binomial probability of finding exactly 2 defective chips.
 - (b) Calculate the approximate Poisson probability of finding exactly 2 defective chips. (What is λ ?)
 - (c) Compare the two answers. Are they close? Why is the Poisson calculation vastly preferred by computers?
4. **Using the Complement Rule with Poisson:** On average, a website goes down $\lambda = 1.5$ times per month.
 - (a) What is the probability that the website goes down **at least once** this month? (Hint: Use the Complement rule $1 - P(0)$).
5. **Expectation and Variance:** Let $Y \sim \text{Pois}(4)$.
 - (a) What is $E[Y]$?
 - (b) What is $\text{Var}(Y)$?
 - (c) What is $E[Y^2]$? (Hint: Remember the variance computational formula $\text{Var}(Y) = E[Y^2] - (E[Y])^2$).

Chapter 12

Continuous Probability Distributions

Discrete distributions (like Binomial and Poisson) are for *counting*. But what happens when we need to measure something that is infinitely divisible, like time, distance, or temperature? We must transition from the discrete mathematics of sums to the continuous mathematics of integrals.

12.1 The Probability Density Function (PDF)

For a continuous random variable X , the probability of hitting any *exact* number is zero. For example, the probability that a person is exactly 175.00000000... cm tall is zero. Therefore, a Probability Mass Function (PMF) cannot exist for continuous variables. Instead, we use a **Probability Density Function (PDF)**.

Definition: Probability Density Function (PDF)

The PDF of a continuous random variable X , denoted $f(x)$, represents the *relative density* of probability at x . To find the actual probability that X falls within a specific interval $[a, b]$, you must calculate the area under the PDF curve between those two points using an integral:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Theorem/Rule: Properties of a Valid PDF

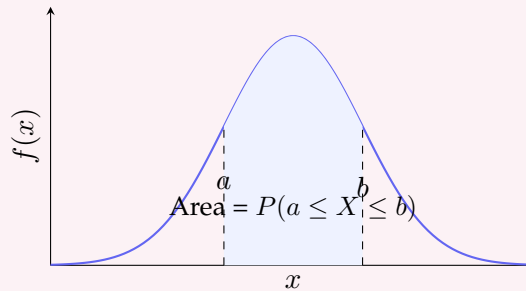
Because probabilities must be valid, a mathematically legal PDF $f(x)$ must satisfy:

1. **Non-negativity:** $f(x) \geq 0$ for all x . (The curve can never dip below the x-axis).
2. **Total Area is 1:** $\int_{-\infty}^{\infty} f(x) dx = 1$. (100% of the probability is under the curve).

Crucial Distinction: Unlike a PMF, the value of $f(x)$ can absolutely be greater than 1. It is a *density*, not a probability. Only the *area* under it must be ≤ 1 .

Core Intuition: Visualizing Probability as Area Under the Density Curve

Below is a Gaussian curve showing how probability $P(a \leq X \leq b)$ corresponds to the shaded area under the curve.



Video Resource: StatQuest: Probability Density Functions (PDFs)



A visually clear explanation of what a PDF is, why $f(x)$ is not a probability, and how to compute probabilities from it using integrals.

12.2 The Uniform Distribution

The simplest continuous distribution is the Uniform Distribution, where probability is spread perfectly evenly across a specific interval.

Definition: Uniform Distribution

A continuous random variable X follows a **Uniform Distribution**, denoted $X \sim \text{Uniform}(a, b)$, if it is equally likely to take any value within the interval $[a, b]$, and has zero probability outside of it.

The PDF is a flat horizontal line:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Theorem/Rule: Expectation and Variance of Uniform

- **Expectation (Mean):** $E[X] = \frac{a+b}{2}$ (The exact midpoint).
- **Variance:** $\text{Var}(X) = \frac{(b-a)^2}{12}$

Data Science Relevance: The Uniform distribution is the foundation of all **Random Number Generation**. Every random library in Python or R starts by generating a number from $\text{Uniform}(0, 1)$ before transforming it into other distributions.

12.3 The Exponential Distribution

While the Poisson distribution counts *how many* rare events happen in a fixed time, the **Exponential distribution** measures the *continuous waiting time* between those events.

Definition: Exponential Distribution

Let X be the waiting time until the next event in a Poisson process with rate λ . $X \sim \text{Exp}(\lambda)$. The PDF is a curve that decays exponentially:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

Theorem/Rule: Expectation and CDF of Exponential

- **Expectation:** $E[X] = \frac{1}{\lambda}$ (If events happen 4 times an hour, the expected wait time is 1/4 of an hour).
- **Variance:** $\text{Var}(X) = \frac{1}{\lambda^2}$
- **Cumulative Distribution Function (CDF):** To avoid integrals, use the CDF to find the probability of waiting *less than or equal to* time x :

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}$$

```

1 import scipy.stats as stats
2 import numpy as np
3
4 # 1. Uniform Distribution Uniform(0, 30)
5 # loc=a, scale=(b-a)
6 uniform_dist = stats.uniform(loc=0, scale=30)
7 print("Uniform Mean:", uniform_dist.mean())
8 print("Uniform P(X > 20):", 1 - uniform_dist.cdf(20))
9
10 # 2. Exponential Distribution Exp(lambda=4 per sec)
11 # scale = 1 / lambda
12 exp_dist = stats.expon(scale=1/4)
13 print("\nExponential Mean wait time:", exp_dist.mean())
14 print("Exponential P(T < 0.5 sec):", exp_dist.cdf(0.5))
15 print("Exponential P(T > 1.0 sec):", 1 - exp_dist.cdf(1.0))

```

Listing 12.1: SciPy/Python Modeling of Uniform and Exponential Distributions

Video Resource: StatQuest: The Exponential Distribution



A deep dive into how the Exponential distribution connects perfectly to the Poisson rate λ .

12.4 Practice Exercises

Detailed step-by-step solutions are available in the Appendix.

1. **PDF Properties:** Let X be a continuous random variable with the PDF: $f(x) = c$ for $0 \leq x \leq 5$, and $f(x) = 0$ everywhere else.
 - (a) Find the value of the constant c that makes this a mathematically valid PDF.
 - (b) What specific continuous distribution is this?
2. **Uniform Distribution:** A train arrives at a station perfectly uniformly between 8:00 AM and 8:30 AM. You arrive at exactly 8:00 AM. Let $X \sim \text{Uniform}(0, 30)$ be your waiting time in minutes.
 - (a) What is the mathematical PDF $f(x)$ for this distribution?
 - (b) What is your Expected waiting time, and what is the Variance?
 - (c) Calculate the exact probability that you wait longer than 20 minutes, $P(X > 20)$. (Use geometry: Area of a rectangle!).
3. **Exponential Mean and Variance:** Call center agents finish calls at an average rate of $\lambda = 0.2$ calls per minute. Let $X \sim \text{Exp}(0.2)$ be the time until the next call is finished.
 - (a) What is the Expected waiting time in minutes?
 - (b) What is the Variance of the waiting time?
4. **Exponential Probabilities (Using CDF):** A server processes database queries at an average rate of $\lambda = 4$ per second. Let T be the time (in seconds) until the next query arrives. $T \sim \text{Exp}(4)$.
 - (a) What is the probability that the next query arrives in *less than* 0.5 seconds? (Hint: Use the CDF $F(0.5)$).
 - (b) What is the probability that it takes *longer than* 1 second for the next query to arrive? (Hint: Use the Complement rule with the CDF).
5. **Data Science Application:** You are building an ML simulation for a logistics company. You need to simulate the time a delivery truck is stuck at a red light. The light cycle is strictly 60 seconds long. Should you model the waiting time using a Poisson, Uniform, or Exponential distribution? Mathematically justify why the other two are wrong.

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Solutions to Practice Exercises

This appendix contains the detailed, step-by-step solutions for all practice exercises found at the end of each weekly chapter.

Week 1: Introduction and Types of Data

Detailed Solutions

1. Concepts: Identify the sample, population, statistic, parameter, descriptive, and inferential parts.

- **Sample:** The 500 patients in the trial.
- **Population:** The general population of all people with high blood pressure.
- **Statistic:** The 12-point average drop in blood pressure (calculated exactly from the sample).
- **Parameter:** The true average drop in blood pressure for the entire population (which is unknown).
- **Descriptive Part:** The act of calculating and reporting the 12-point drop in the 500 patients.
- **Inferential Part:** The act of generalizing that result to conclude the drug works for everyone.

2. Data Classification: Classify the variables.

- **(a) Emails received:** Numerical Discrete (you can count them: 0, 1, 2, but you can't get 1.5 emails).
- **(b) Letter grades (A, B, C, D, F):** Categorical Ordinal (they are categories, but A is definitively better than B).
- **(c) Exact volume of water:** Numerical Continuous (it can be exactly 500.1234 ml, falling anywhere on the interval).
- **(d) Laptop brand:** Categorical Nominal (categories with no natural rank or order).

3. Scales of Measurement: Identify the scale.

- **(a) Manufacturing_Year: Interval.** Gaps are equal ($2018 - 2015 = 3$ years), but there is no absolute zero (Year 0 doesn't mean "no time").
- **(b) Mileage_Driven: Ratio.** A car with 0 km has truly been driven "zero distance."
- **(c) Color: Nominal.** Unordered labels.
- **(d) Condition_Rating (1-Poor, 2-Fair, 3-Excellent): Ordinal.** The numbers imply order, but the gap between Poor and Fair isn't mathematically identical to Fair and Excellent.

4. Data Structures: Time Series vs. Cross-Sectional.

- **Patient X over a week: Time Series.** It observes a single subject across multiple time points.
- **100 patients on Monday morning: Cross-Sectional.** It observes multiple subjects at a single snapshot in time.

5. Data Science Application: Why is treating Pincode as Numerical Ratio a catastrophe?

- A Pincode is a **Categorical Nominal** variable that just happens to use numbers as its labels.
- If you treat it as Numerical Ratio, the machine learning model will try to do math on it. It will assume that Pincode 600000 is "twice as large" or "mathematically greater" than Pincode 300000.
- This implies geographic locations have a numerical order or magnitude, which is complete nonsense and will destroy the model's logic. Pincodes should be treated as categories.

Week 2: Describing Categorical Data**Detailed Solutions****1. Frequency Calculations: Calculate the relative frequency and percentage.**

- The total sample size is $n = 200$.
- Android frequency = 110. Relative frequency = $110/200 = 0.55$ (55%).
- iOS frequency = 80. Relative frequency = $80/200 = 0.40$ (40%).
- Other frequency = $200 - 110 - 80 = 10$. Relative frequency = $10/200 = 0.05$ (5%).

2. Charting Logic: Which chart to convey exactly 50%?

- A **Pie Chart** is actually better suited here.
- While humans are generally poor at estimating obscure angles (like 17%), a 50% chunk perfectly cuts a circle in half. This "half-pie" visual is instantly and universally recognized by audiences without needing to read the axis labels of a Bar Chart.

3. Graphing Ethics: What cardinal rule did they violate?

- They violated the rule: **"Always start the y-axis at zero for bar charts."**
- They likely started the y-axis at 9,000. This made the competitor's bar (10,000) visually appear to have a height of 1,000 units above the baseline, and the company's bar (11,000) visually appear to have a height of 2,000 units above the baseline. Thus, it falsely looks twice as tall.

4. Mode and Median: Small (10), Medium (20), Large (15), Extra Large (5).

- **Mode:** The category with the highest frequency is **Medium** (20).
- **Median:** The categories are ordinal and naturally sorted (S, M, L, XL). Total $n = 50$, an even number. The median lies between the 25th and 26th observations.
- Cumulative counts: Small (10), Medium ($10 + 20 = 30$).
- Since both the 25th and 26th observations fall within the Medium category, the **Median is Medium**.

5. Data Science Application: What is the accuracy of the terrible model?

- The model guesses "Benign" for all 10,000 images.
- It gets all 9,800 Benign images correct, and gets the 200 Malignant images wrong.
- **Accuracy** = $9,800/10,000 = 0.98$ (**98%**).
- This specific dataset problem is called **Class Imbalance**. It proves that raw accuracy is a terrible metric when predicting rare, critical events!

Week 3: Describing Numerical Data**Detailed Solutions****1. Central Tendency: Startup Salaries**

- **(a) Mean:** $\bar{x} = \frac{80000 \times 4 + 1680000}{5} = \frac{2000000}{5} = 400,000$. **Median:** Sorted array is 80k, 80k, 80k, 80k, 1.68M. The middle value is exactly 80,000.
- **(b)** The Mean is heavily distorted by the single extreme outlier (the CEO). Quoting the Mean is mathematically true but highly deceptive, as 80% of the company makes $5\times$ less than that average. Because the extreme outlier is on the high (right) end, making Mean > Median, the distribution is **Right-Skewed**.

2. Calculating Spread: $X = \{2, 4, 4, 4, 5, 5, 7, 9\}$

- **(a) Sample Mean ($\bar{x}$):** $\frac{2+4+4+4+5+5+7+9}{8} = \frac{40}{8} = 5$.
- **(b) Sample Variance (s^2):**
 1. Deviations from mean ($x_i - 5$): $-3, -1, -1, -1, 0, 0, 2, 4$.
 2. Squared deviations: $9, 1, 1, 1, 0, 0, 4, 16$. Sum of squares = 32.
 3. Divide by $n - 1$: $s^2 = \frac{32}{7} \approx 4.57$.
- **Standard Deviation (s):** $s = \sqrt{4.57} \approx 2.14$.

3. Quartiles and IQR: $X = \{2, 4, 4, 4, | 5, 5, 7, 9\}$

- **(a) Median (Q_2):** The dataset has $n = 8$. The median is the average of the 4th and 5th values: $\frac{4+5}{2} = 4.5$.
- **(b) Q_1 :** The median of the lower half $\{2, 4, 4, 4\}$ is $\frac{4+4}{2} = 4$. **Q_3 :** The median of the upper half $\{5, 5, 7, 9\}$ is $\frac{5+7}{2} = 6$.
- **(c) IQR:** $Q_3 - Q_1 = 6 - 4 = 2$.

4. Data Science Application: Feature Scaling

- The measures of dispersion (Range, Variance, Standard Deviation) for Square Footage are thousands of times larger than those for Bedrooms.
- Without scaling, a distance-based ML model will treat a change of 1 sq ft as mathematically equal in importance to a change of 1 bedroom. Because square footage varies by thousands, it will completely dominate the loss function, and the algorithm will functionally ignore the Bedrooms feature entirely.

5. Histograms vs. Bar Charts

- **Visual Difference:** Bar charts have gaps between the bars; Histograms have adjacent bars that touch.
- **Underlying Reason:** Bar charts display *Categorical* data, where groups are distinct and disjoint (there is no logical space "between" iOS and Android). Histograms display *Continuous Numerical* data along a continuous number line; the exact end of one bin (e.g., 10 – 20) is the exact start of the next bin (20 – 30), hence they touch.

Week 4: Association Between Two Variables

Detailed Solutions

1. Contingency Tables: Housing vs. Commute

- (a) **Table Construction:**

	On-Campus	Off-Campus	Total
Walk	80	20	100
Bike	20	40	60
Car	10	130	140
Total	110	190	300

- (b) **Conditional on On-Campus:** Total On-Campus = 110.
Walk = $80/110 \approx 72.7\%$. Bike = $20/110 \approx 18.2\%$. Car = $10/110 \approx 9.1\%$.
- (c) **Association:** Yes, they are highly associated. We prove this by looking at the Off-Campus conditional distribution (Car = $130/190 \approx 68.4\%$). Because the commute distributions (9.1% Car vs 68.4% Car) change drastically depending on housing, the variables are dependent/associated.

2. Correlation Calculation ($X = \{1, 2, 3, 4, 5\}$, $Y = \{2, 4, 5, 4, 5\}$)

- $\bar{x} = 3, \bar{y} = 4$.
- Deviations ($x_i - \bar{x}$): $-2, -1, 0, 1, 2$.
- Deviations ($y_i - \bar{y}$): $-2, 0, 1, 0, 1$.
- Cross-products $\sum (x_i - \bar{x})(y_i - \bar{y}) = (-2)(-2) + (-1)(0) + (0)(1) + (1)(0) + (2)(1) = 4 + 0 + 0 + 0 + 2 = 6$.
- Sum of squares X : $4 + 1 + 0 + 1 + 4 = 10$.
- Sum of squares Y : $4 + 0 + 1 + 0 + 1 = 6$.
- $r = \frac{6}{\sqrt{10} \cdot \sqrt{6}} = \frac{6}{\sqrt{60}} \approx \frac{6}{7.746} \approx 0.775$.

3. Correlation Fallacies: Ice Cream and Sharks

- (a) $r = +0.85$ indicates a **strong, positive, linear** relationship. As ice cream sales increase, shark attacks tightly increase.
- (b) The fallacy is "**Correlation implies Causation**". The confounding variable is **Summer (Temperature)**. When it gets hot, ice cream sales go up. Simultaneously, more people swim in the ocean, causing shark attacks to go up. One does not cause the other.

4. Data Science Application: Mileage Features

- (a) $r = 1.0$. Miles is just Kilometers multiplied by a constant (0.62). It is a perfect positive linear transformation.
- (b) No, you must drop one. Including both provides zero new information to the model and causes a mathematical instability called **Multicollinearity**.

5. Non-Linear Relationships ($X = \{-2, -1, 0, 1, 2\}$, $Y = \{4, 1, 0, 1, 4\}$)

- (a) The relationship is exactly $Y = X^2$. It is a perfect parabola.
- (b) $r = 0$. Pearson's r only measures *linear* relationships. A straight line of best fit through a U-shaped parabola is completely flat, returning $r = 0$, completely failing to detect the perfect non-linear relationship.

Week 5: Permutations and Combinations

Detailed Solutions

1. Basic Counting: Burger Menu

- We cascade the tasks using the Multiplication Principle (AND rule).
- Pick a bun (3 ways) AND pick a meat (4 ways) AND pick a side (Fries OR Rings = 2 ways).
- $3 \times 4 \times 2 = 24$ distinct meals.

2. Permutations (Distinct): 8 Runners

- (a) ${}^8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$ ways.
- (b) We used the **Permutation** formula because the *order* of selection fundamentally matters. (Alice coming 1st and Bob 2nd is a totally different outcome than Bob coming 1st and Alice 2nd).

3. Permutations (Non-Distinct): "STATISTICS"

- (a) The word has 10 total letters. The frequencies are: S (3), T (3), I (2), A (1), C (1).
- Number of unique arrangements = $\frac{10!}{3! \times 3! \times 2!} = \frac{3,628,800}{6 \times 6 \times 2} = \frac{3,628,800}{72} = 50,400$.
- (b) If we just used $10!$, the math assumes every 'S' is a completely unique letter (like S_1, S_2, S_3). We divide by $3!$ in the denominator to mathematically erase the 6 internal ways the three 'S's can swap positions with each other without changing the spelling of the word.

4. Combinations: ML Models

- (a) Order does not matter for an ensemble group. ${}^nC_r = {}^{10}C_4 = \frac{10!}{4!(10-4)!} = \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} = 210$ combinations.
- (b) ${}^{10}C_6 = \frac{10!}{6!4!} = 210$. The number is identical due to the symmetry rule of combinations (${}^nC_r = {}^nC_{n-r}$). Logically, choosing exactly 4 models to keep is mathematically identical to choosing exactly 6 models to throw in the trash.

5. Advanced Application: Committees

- (a) No restrictions: Total pool is 13 people ($6M + 7W$). Pick 5.
 ${}^{13}C_5 = \frac{13!}{5!8!} = 1,287$.

- **(b) Exactly 3 Men AND 2 Women:**
 - Ways to pick the Men: ${}^6C_3 = \frac{6!}{3!3!} = 20$.
 - Ways to pick the Women: ${}^7C_2 = \frac{7!}{2!5!} = 21$.
 - Multiply them together (AND rule): $20 \times 21 = 420$ committees.

Week 6: Probability Basics

Detailed Solutions

1. Sample Spaces: Define S .

- **(a) Flipping a coin 3 times:** $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. (Discrete, finite, 8 elements).
- **(b) Counting cars in an hour:** $S = \{0, 1, 2, 3, 4, \dots\}$. (Discrete, theoretically countably infinite).
- **(c) Lifespan of a lightbulb:** $S = \{t \mid t \geq 0\}$. (Continuous interval from 0 to infinity).

2. Set Operations: $S = \{1, 2, 3, 4, 5, 6\}$

- **(a) Sets:** $A = \{2, 4, 6\}$, $B = \{4, 5, 6\}$.
- **(b) $A \cap B$:** $\{4, 6\}$. In plain English: "Rolling an even number that is also greater than 3."
- **(c) $A \cup B$:** $\{2, 4, 5, 6\}$.
- **(d) A^c :** $\{1, 3, 5\}$. (The event of rolling an odd number).
- **(e) Mutually Exclusive? No.** For them to be mutually exclusive, $A \cap B$ must be the empty set ($\emptyset$). Because they share the outcomes 4 and 6, they can happen at the same time.

3. Complement Rule: Cybersecurity Packets

- **(a)** To calculate $P(\text{at least one})$ directly using the Addition Rule, you would have to calculate $P(1) + P(2) + P(3) + P(4) + P(5)$. This involves heavy combinatorics for every single scenario.
- **(b)** The complement of "at least one packet drops" is "**zero packets drop**" (i.e., all 5 packets succeed).
- Probability of one packet succeeding = $1 - 0.1 = 0.9$.
- Probability of all 5 succeeding independently = $0.9^5 = 0.59049$.
- By the Complement Rule: $P(\text{at least one drops}) = 1 - 0.59049 = 0.40951$.

4. General Addition Rule: Students studying Python and R Let $P(P) = 0.60$, $P(R) = 0.50$, and $P(P \cap R) = 0.20$.

- **(a) Python OR R ($P \cup R$):**

$$P(P \cup R) = P(P) + P(R) - P(P \cap R) = 0.60 + 0.50 - 0.20 = 0.90 \text{ (90\%)}$$

- **(b) Neither:** "Neither" is the complement of "Python OR R".

$$P((P \cup R)^c) = 1 - 0.90 = 0.10 \text{ (10\%)}$$

5. Axioms Check: Machine Learning Probabilities

- The model outputs sum to: $0.5 + 0.4 + 0.2 = 1.1$.
- This fundamentally violates **Kolmogorov's Second Axiom (Normalization)**, which mathematically mandates that the probability of the entire sample space must equal exactly 1.0 ($P(S) = 1$).

Week 7: Conditional Probability and Bayes' Theorem**Detailed Solutions****1. Conditional Probability: Deck of Cards**

- (a) $P(K) = \frac{4}{52} = \frac{1}{13}$.
- (b) The new information restricts our sample space from 52 down to just 12 face cards. Out of those 12 Face cards, 4 are Kings. $P(K | F) = \frac{4}{12} = \frac{1}{3}$. (Notice how the new information drastically changed the probability!)

2. Multiplication Rule: Marbles without Replacement

- The bag starts with 8 total marbles.
- $P(R_1) = \frac{5}{8}$.
- Once the Red marble is removed, the bag has 7 marbles left, and 3 of them are Blue. $P(B_2 | R_1) = \frac{3}{7}$.
- By the Multiplication Rule: $P(R_1 \cap B_2) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$.

3. Checking for Independence

- (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

$$0.7 = 0.4 + 0.5 - P(A \cap B) \implies P(A \cap B) = 0.9 - 0.7 = \mathbf{0.2}$$

- (b) To test for independence, we check if $P(A \cap B) = P(A) \times P(B)$.

$$P(A) \times P(B) = 0.4 \times 0.5 = 0.2$$

Since $0.2 = 0.2$, the events **ARE** mathematically independent.

4. Bayes' Theorem: Medical Testing

- (a) **Total $P(+)$:** A positive test can come from a sick person (True Positive) OR a healthy person (False Positive).

$$P(+) = P(D)P(+ | D) + P(H)P(+ | H)$$

$$P(+) = (0.01 \times 0.90) + (0.99 \times 0.05) = 0.009 + 0.0495 = \mathbf{0.0585}$$

- (b) $P(D | +)$: Using Bayes' Theorem:

$$P(D | +) = \frac{P(D)P(+ | D)}{P(+)} = \frac{0.009}{0.0585} \approx 0.1538 \text{ (15.38\%)}$$

Even with a "90% accurate" test, a positive result only means a 15% chance of having the disease! This is the Base Rate Fallacy.

5. Data Science Application: Spam Filter

- Let $S = \text{Spam}$ (0.20), $L = \text{Legit}$ (0.80). Let $F = \text{contains the word "Free"}$.
- We know $P(F | S) = 0.80$ and $P(F | L) = 0.10$.
- First find total probability of "Free":

$$P(F) = (0.20 \times 0.80) + (0.80 \times 0.10) = 0.16 + 0.08 = 0.24$$

- Now apply Bayes' Theorem to find $P(S | F)$:

$$P(S | F) = \frac{0.20 \times 0.80}{0.24} = \frac{0.16}{0.24} = \frac{2}{3} \approx \mathbf{66.67\%}$$

Week 8: Random Variables**Detailed Solutions****1. Discrete vs. Continuous**

- (a) Complaints: **Discrete**. You count them (0, 1, 2). You can't get 1.5 complaints.
- (b) Load time: **Continuous**. Time is an infinitely dense measurement (e.g., 1.245 seconds).
- (c) Heads: **Discrete**. You count them (0 to 100).
- (d) Volume: **Continuous**. Liquid is measured, not counted.

2. Validating a PMF

- (a) By the sum rule, $\sum P(X = x) = 1$.

$$c(1) + c(2) + c(3) + c(4) = 1 \implies 10c = 1 \implies \mathbf{c = 0.1}$$

- (b) $P(X = 3) = 0.1 \times 3 = \mathbf{0.3}$.

3. Creating a PMF: Two Dice

- (a) The sample space $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$. The size is $6 \times 6 = \mathbf{36}$.
- (b) The numerical values X (the sum) can take: $\{\mathbf{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12}\}$.
- (c) For $X = 7$, the winning outcomes are (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) (there are 6).
 $P(X = 7) = \frac{6}{36} = \frac{1}{6}$.
- For $X = 12$, the only winning outcome is (6, 6). $P(X = 12) = \frac{1}{36}$.

4. Building a CDF

- (a) **CDF Values:**

- $F(1) = P(X \leq 1) = P(X = 1) = \mathbf{0.2}$.
- $F(2) = P(X \leq 2) = P(X = 1) + P(X = 2) = 0.2 + 0.5 = \mathbf{0.7}$.
- $F(3) = P(X \leq 3) = 0.7 + 0.3 = \mathbf{1.0}$.

- (b) $F(1.5)$: What is the probability $X \leq 1.5$? Because X can only be 1, 2, or 3, the only possible value less than 1.5 is exactly 1. So $F(1.5) = F(1) = \mathbf{0.2}$.

- **(c) Graph:** It is a **Step Function**. It stays flat at 0, jumps up to 0.2 exactly at $x = 1$, stays flat at 0.2, jumps up to 0.7 exactly at $x = 2$, stays flat, jumps up to 1.0 exactly at $x = 3$, and stays flat at 1.0 forever to the right.

5. Using the CDF

- The CDF accumulates probability. To find the exact probability mass added at $x = 3$, we take the total accumulated probability up to 3, and subtract the total accumulated probability up to 2.
- $P(Y = 3) = P(Y \leq 3) - P(Y \leq 2) = F(3) - F(2) = 0.9 - 0.4 = \mathbf{0.5}$.

Week 9: Expectation and Variance

Detailed Solutions

1. Calculating Expectation: Lottery Ticket

- **(a) PMF:** $P(X = 500) = 0.01$, $P(X = 50) = 0.05$, $P(X = 0) = 0.94$.
- **(b) $E[X]$:** Multiply each payout by its probability and sum.

$$E[X] = (500 \times 0.01) + (50 \times 0.05) + (0 \times 0.94) = 5 + 2.5 + 0 = \mathbf{7.5 \text{ Rupees}}$$

- **(c) Net Profit:** You pay 10 to play. The Expected Net Profit is $E[X - 10] = E[X] - 10 = 7.5 - 10 = \mathbf{-2.5 \text{ Rupees}}$. Mathematically, you are guaranteed to lose money in the long run.

2. Calculating Variance

- **(a) $E[Y]$:**

$$E[Y] = (-1 \times 0.2) + (0 \times 0.5) + (1 \times 0.3) = -0.2 + 0 + 0.3 = \mathbf{0.1}$$

- **(b) $E[Y^2]$:** Square the values first, then multiply by the original PMF probabilities.

$$E[Y^2] = ((-1)^2 \times 0.2) + (0^2 \times 0.5) + (1^2 \times 0.3) = (1 \times 0.2) + 0 + (1 \times 0.3) = \mathbf{0.5}$$

- **(c) $Var(Y)$:** Using the computational formula $E[Y^2] - (E[Y])^2$:

$$Var(Y) = 0.5 - (0.1)^2 = 0.5 - 0.01 = \mathbf{0.49}$$

3. Properties of Expectation and Variance Given $E[X] = 25$, $Var(X) = 4$, and $F = 1.8X + 32$.

- **(a) $E[F]$:** Expectation is linear.

$$E[1.8X + 32] = 1.8E[X] + 32 = 1.8(25) + 32 = 45 + 32 = \mathbf{77^\circ F}$$

- **(b) $Var(F)$:** Variance ignores shifts (+32 disappears) and squares scalars (1.8 becomes 1.8^2).

$$Var(1.8X + 32) = 1.8^2 Var(X) = 3.24 \times 4 = \mathbf{12.96}$$

- **(c) $SD(F)$:** $\sqrt{Var(F)} = \sqrt{12.96} = \mathbf{3.6^\circ F}$. (Notice that $1.8 \times \sqrt{4} = 3.6$; standard deviation does not square the scalar!).

4. Independence and Variance

- **(a) $E[X+Y]$:** By the strict Linearity of Expectation, $E[X+Y] = E[X] + E[Y] = 0.5 + 3.5 = \mathbf{4.0}$. This is always true, even if they were dependent.
- **(b)** Flipping a coin and rolling a die have absolutely no physical connection, so they are perfectly **independent**. Because they are independent, we can legally use the additive property of variance: $Var(X + Y) = Var(X) + Var(Y)$.

Week 10: The Binomial Distribution

Detailed Solutions

1. Bernoulli Trials: CTR

- **(a)** The distribution is exactly $X \sim \text{Bernoulli}(p = 0.03)$.
- **(b)** Expected Value $E[X] = p = \mathbf{0.03}$. Variance $\text{Var}(X) = p(1 - p) = 0.03 \times 0.97 = \mathbf{0.0291}$.

2. Binomial PMF: Defective Bulbs

- **(a)** $n = 5, p = 0.10$.
- **(b)** $P(X = 2)$: Using the Binomial PMF formula:

$$P(X = 2) = \binom{5}{2} (0.10)^2 (0.90)^3 = 10 \times 0.01 \times 0.729 = \mathbf{0.0729} \text{ (7.29\%)}$$

- **(c)** $P(X = 0)$:

$$P(X = 0) = \binom{5}{0} (0.10)^0 (0.90)^5 = 1 \times 1 \times (0.90)^5 = \mathbf{0.59049} \text{ (59.05\%)}$$

3. Binomial Expectation and Variance: Multiple Choice

- We have $n = 50$ trials. The probability of guessing correctly is $p = 1/4 = 0.25$.
- **(a) Expected Score:** $E[Y] = n \times p = 50 \times 0.25 = \mathbf{12.5}$ questions.
- **(b) Variance:** $\text{Var}(Y) = n \times p \times (1 - p) = 50 \times 0.25 \times 0.75 = \mathbf{9.375}$.

4. Using the Complement Rule with Binomial

- We have $n = 10, p = 0.60$. We want $P(X \geq 1)$.
- Calculating $P(1) + P(2) + \dots + P(10)$ is agonizing. Instead, we use the Complement Rule: $P(X \geq 1) = 1 - P(X = 0)$.
- $P(X = 0) = \binom{10}{0} (0.60)^0 (0.40)^{10} = (0.40)^{10} \approx 0.0001048$.
- $P(X \geq 1) = 1 - 0.0001048 = \mathbf{0.9998952}$. There is a 99.99% chance at least one person is cured.

5. Data Science Application: A/B Testing

- **Assumption 1 (Independent):** Users do not communicate or influence each other's decisions to click.
- **Assumption 2 (Identically Distributed):** Every single user in the 100-person sample has the exact same baseline underlying probability of clicking.
- **Violation Example:** If a user posts the link in a Discord server saying "Click this button for free money!", you suddenly get a flood of highly-motivated users. Their baseline probability of clicking is vastly higher than the organic users, violently breaking the "Identically Distributed" assumption and invalidating your entire Binomial model.

Week 11: The Poisson Distribution

Detailed Solutions

1. Poisson Probabilities: Bookstore

- (a) $P(X = 3)$: Here, $\lambda = 2$.

$$P(X = 3) = \frac{e^{-2}2^3}{3!} \approx \frac{0.1353 \times 8}{6} = \mathbf{0.1804} \text{ (18.04\%)}$$

- (b) $P(X = 0)$:

$$P(X = 0) = \frac{e^{-2}2^0}{0!} = \frac{e^{-2} \times 1}{1} \approx \mathbf{0.1353} \text{ (13.53\%)}$$

2. Scaling the Interval: DDoS Requests

- (a) **2-minute window**: The base rate is 6 per minute. For 2 minutes, the new rate is $\lambda_{\text{new}} = 6 \times 2 = 12$.

$$P(X = 10) = \frac{e^{-12}12^{10}}{10!} \approx \mathbf{0.1048} \text{ (10.48\%)}$$

- (b) **10-second window**: 10 seconds is $\frac{1}{6}$ of a minute. The new rate is $\lambda_{\text{new}} = 6 \times \frac{1}{6} = 1$. Because $E[X] = \lambda$, the expected number of requests is exactly 1.

3. Poisson Approximation to Binomial

- (a) **Exact Binomial** ($n = 100, p = 0.01$):

$$P(X = 2) = \binom{100}{2} (0.01)^2 (0.99)^{98} = 4950 \times 0.0001 \times 0.3733 \approx \mathbf{0.1849}$$

- (b) **Poisson Approx** ($\lambda = np = 100 \times 0.01 = 1$):

$$P(X = 2) = \frac{e^{-1}1^2}{2!} = \frac{0.3678}{2} = \mathbf{0.1839}$$

- (c) **Comparison**: The numbers are extremely close (a 0.001 difference). We strongly prefer the Poisson distribution because computing $100!$ inherently causes massive memory integer overflows on normal computers, while calculating e^{-1} is computationally trivial.

4. Using the Complement Rule with Poisson

- We want $P(X \geq 1)$. The complement is $P(X = 0)$.

$$P(X = 0) = \frac{e^{-1.5}1.5^0}{0!} = e^{-1.5} \approx \mathbf{0.2231}.$$

$$P(X \geq 1) = 1 - 0.2231 = \mathbf{0.7769} \text{ (77.69\%)}.$$

5. Expectation and Variance: $Y \sim \text{Pois}(4)$

- (a) $E[Y] = \lambda = 4$.
- (b) $\text{Var}(Y) = \lambda = 4$.
- (c) Using the variance formula $\text{Var}(Y) = E[Y^2] - (E[Y])^2$:

$$4 = E[Y^2] - (4)^2 \implies 4 = E[Y^2] - 16 \implies E[Y^2] = \mathbf{20}$$

Week 12: Continuous Probability Distributions

Detailed Solutions

1. PDF Properties

- (a) The total area under the PDF must equal 1. The shape is a rectangle from $x = 0$ to $x = 5$, so the width is 5. The height is c .

$$\text{Area} = \text{width} \times \text{height} = 5 \times c = 1 \implies c = \frac{1}{5} = 0.2$$

- (b) Because the probability is perfectly flat across an interval, this is the **Uniform Distribution**, specifically $X \sim \text{Uniform}(0, 5)$.

2. Uniform Distribution: Waiting for a Train

- (a) **Mathematical PDF:** $f(x) = \frac{1}{30}$ for $0 \leq x \leq 30$.
- (b) **Expectation and Variance:**

$$E[X] = \frac{0 + 30}{2} = 15 \text{ minutes}$$

$$\text{Var}(X) = \frac{(30 - 0)^2}{12} = \frac{900}{12} = 75$$

- (c) $P(X > 20)$: We need the area under the PDF from 20 to 30. This is a rectangle with width $(30 - 20) = 10$, and height $1/30$.

$$\text{Area} = 10 \times \frac{1}{30} = \frac{1}{3} \approx 33.33\%$$

3. Exponential Mean and Variance

- Here, $\lambda = 0.2$.
- (a) **Expected wait time:** $E[X] = \frac{1}{\lambda} = \frac{1}{0.2} = 5$ minutes.
- (b) **Variance:** $\text{Var}(X) = \frac{1}{\lambda^2} = \frac{1}{0.04} = 25$.

4. Exponential Probabilities: Database Queries

- Here, $\lambda = 4$.
- (a) $P(T < 0.5)$: We use the CDF $F(x) = 1 - e^{-\lambda x}$.

$$F(0.5) = 1 - e^{-4(0.5)} = 1 - e^{-2} \approx 1 - 0.1353 = 0.8647 \text{ (86.47\%)}$$

- (b) $P(T > 1)$: Use the complement rule $1 - P(T \leq 1)$.

$$P(T > 1) = 1 - F(1) = 1 - (1 - e^{-4(1)}) = e^{-4} \approx 0.0183 \text{ (1.83\%)}$$

5. Data Science Application: Simulation

- You should use the **Uniform Distribution**, specifically $\text{Uniform}(0, 60)$. Assuming you arrive at a perfectly random time within the cycle, your wait time is equally likely to be anywhere between 0 and 60 seconds.
- **Why not Poisson?** Poisson models discrete counts (number of events), but waiting time is continuous.
- **Why not Exponential?** The Exponential distribution mathematically extends to infinity ($x \geq 0$). A red light has a strict 60-second cutoff, so the unbounded tail of the Exponential distribution makes it mathematically impossible and inappropriate here.